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Symplectic Groupoids and Physical Applications

Bachelor's thesis
in the field of Physics

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Summary

This Bachelor's thesis gives an introductory account of Lie groupoids, Lie algebroids, and symplectic groupoids, with emphasis on their role in geometric mechanics and mathematical physics. We provide the basic definitions and examples of Lie groupoids and Lie algebroids, Poisson manifolds and the cotangent bundle of a Poisson manifold is a Lie algebroid. Then we study symplectic groupoids as global integrations of Poisson manifolds. Finally, two elementary physical examples are discussed: the pair groupoid of an ordinary symplectic phase space and the cotangent groupoid $T^*G \rightrightarrows \mathfrak{g}^*$, which integrates the Lie–Poisson structure appearing in rigid body dynamics and systems with internal symmetry. The aim is not to give a complete treatment of symplectic groupoids, but to show how they provide a natural global language for Poisson phase spaces, Hamiltonian dynamics, symmetries, momentum maps and possible quantisation procedures.

Keywords

Lie groupoids, Lie algebroids, Poisson geometry, symplectic groupoids, Hamiltonian mechanics

Title of the thesis in Polish language

Grupoidy symplektyczne i zastosowania fizyczne

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List of symbols

M, N :	smooth finite-dimensional manifolds,
G :	Lie group,
$\mathfrak{g}$:	Lie algebra of G ,
$\mathfrak{g}^*$:	dual vector space of $\mathfrak{g}$,
$\mathcal{G} \rightrightarrows M$:	Lie groupoid over M ,
s, t :	source and target maps of a groupoid,
$\mathcal{G}_2$:	manifold of composable pairs of $\mathcal{G}$,
$A(\mathcal{G})$:	Lie algebroid of a Lie groupoid $\mathcal{G}$,
ρ :	anchor map of a Lie algebroid,
TM, T^*M :	tangent and cotangent bundles of M ,
$\mathfrak{X}(M)$:	space of vector fields on M ,
$\Omega^k(M)$:	space of differential k -forms on M ,
π :	Poisson bivector on M ,
$\pi^\sharp$:	bundle map $T^*M \rightarrow TM$ induced by π ,
$\{\cdot, \cdot\}$:	Poisson bracket of functions,
ω :	symplectic form on a phase space,
Ω :	symplectic form on a symplectic groupoid,
$\overline{\mathcal{G}}$:	same manifold as $\mathcal{G}$ with symplectic form $-\Omega$,
X_H :	Hamiltonian vector field of a Hamiltonian H ,
μ :	element of $\mathfrak{g}^*$ or a momentum map, according to context.

Introduction

The phase space of a conservative mechanical system is usually modelled by a symplectic manifold (M, ω) . A Hamiltonian $H \in C^\infty(M)$ determines a vector field X_H through

$$\iota_{X_H} \omega = dH,$$

and the integral curves of X_H are the trajectories of the system. The symplectic form also equips the observables with the Poisson bracket $\{f, g\} = \omega(X_f, X_g)$, in terms of which the equations of motion read $\dot{f} = \{f, H\}$.

This description is not general. Reduction by a symmetry group, for instance the passage to the body frame of a rigid body, produces a smaller phase space that carries a Poisson bracket but, in general, no symplectic form. Such a space is a *Poisson manifold*. Every Hamiltonian still determines a vector field $X_H = \{\cdot, H\}$, but the manifold decomposes into *symplectic leaves*, submanifolds of possibly varying dimension on which the bracket is non-degenerate, and every trajectory remains in the leaf through its initial condition. The standard example is the free rigid body: its reduced phase space is the dual $\mathfrak{so}(3)^*$ of the rotation algebra, the symplectic leaves are the spheres of constant angular momentum, and Euler's equations describe the Hamiltonian flow on them.

The subject of this thesis is the *symplectic groupoid*: a manifold that is simultaneously a Lie groupoid over a given manifold M and a symplectic manifold, with the two structures related by a single compatibility condition. The base of a symplectic groupoid carries a canonical Poisson structure, and a Poisson manifold that is integrable at all is integrated by such a groupoid. The guiding analogy is

$$\text{symplectic groupoid} : \text{Poisson manifold} = \text{Lie group} : \text{Lie algebra}.$$

A Lie group determines its Lie algebra by linearisation at the identity; a symplectic groupoid determines a Poisson manifold by restriction to its units.

Symplectic groupoids were introduced in the 1980s, independently, by Karasev [11], Weinstein [9] and Zakrzewski [12], in all three cases with quantisation as the motivation. The systematic theory was developed by Coste, Dazord and Weinstein [10], and the integrability problem, that is, the question of which Poisson manifolds admit a symplectic groupoid, was solved by Crainic and Fernandes in 2004 [14]. More recently, symplectic groupoids have been used in numerical analysis as the geometric basis of structure-preserving integrators for Poisson systems [18, 27].

The two sides of the analogy are connected by an intermediate object, the *Lie algebroid*, which is the infinitesimal model of a Lie groupoid. Every Poisson manifold (M, π) carries a canonical Lie algebroid structure on its cotangent bundle T^*M , with anchor and bracket determined by π , and a symplectic groupoid of (M, π) is a Lie groupoid integrating this cotangent algebroid. The thesis follows the chain

$$\begin{aligned} \text{Hamiltonian mechanics} &\rightsquigarrow \text{Poisson manifolds} \rightsquigarrow \\ T^*M \text{ as a Lie algebroid} &\rightsquigarrow \text{symplectic groupoids}, \end{aligned}$$

one step per chapter.

The thesis is expository. It assumes the differential geometry, Lie theory and classical mechanics of an undergraduate course. The material of Chapters 1–3 is standard and follows mainly [4] and [6]; the mechanics follows [8]. The main theorems of Chapter 3 are proved in full; results quoted without proof carry a reference. In particular, the integrability theorem of Crainic and Fernandes is stated and used, not proved, although the obstruction is illustrated by an example of a Poisson manifold that admits no symplectic groupoid. Quantisation is only indicated in the Conclusion.

Chapter 1 introduces Lie groupoids and Lie algebroids, with three running examples: a Lie group, the pair groupoid $M \times M$ of a manifold, and the action groupoid of a group action. Chapter 2 develops Poisson geometry: Hamiltonian vector fields, Casimir functions and the symplectic foliation; the Lie–Poisson structure on $\mathfrak{g}^*$, momentum maps and the rigid body; and the cotangent Lie algebroid of a Poisson manifold. Chapter 3 defines symplectic groupoids, derives their structural properties, proves that the base of a symplectic groupoid is a Poisson manifold and that the Lie algebroid of the groupoid is the cotangent algebroid of the base, and discusses integrability. Chapter 4 contains the physical applications: canonical transformations and Hamiltonian flows on the pair groupoid, Lie–Poisson reduction of the rigid body on the cotangent groupoid $T^*G \rightrightarrows \mathfrak{g}^*$, and Poisson integrators.

Sign conventions in Poisson geometry differ between sources; ours are fixed where each object first appears. A list of symbols precedes this introduction.

Chapter 1

Lie Groupoids and Lie Algebroids

1.1. From groups to groupoids

Groups are the standard tool to describe symmetry in geometry. A space being highly symmetric means that its symmetry group is large and acts on it transitively. However, many objects that are symmetric intuitively have only a small symmetry group. The following example of Weinstein [13] shows the problem and its solution.

Consider the tiling $\mathcal{T}$ of the plane by 2×1 rectangles, with edge set

$$\mathcal{T} = (2\mathbb{Z} \times \mathbb{R}) \cup (\mathbb{R} \times \mathbb{Z}) \subset \mathbb{R}^2.$$

Its symmetry group $G_{\mathcal{T}}$, the subgroup of the Euclidean group $E(2)$ which preserves $\mathcal{T}$, contains the translations by the lattice $2\mathbb{Z} \times \mathbb{Z}$ and several types of reflections. Consider now a finite region $B = [0, 2m] \times [0, n]$. The symmetry group of $\mathcal{T} \cap B$ is just the Klein four-group generated by the two reflections through the centre of B , independently of m and n . It consists of the transformations that preserve the whole pattern but it says nothing about the local symmetries which relate individual points.

To record such information, instead of a transformation φ alone, we consider the set of the triples

$$\mathcal{G}_{\mathcal{T}} = \{ (y, \varphi, x) : \varphi \in G_{\mathcal{T}}, x, y \in \mathbb{R}^2, y = \varphi(x) \}.$$

Two triples (z, ψ, y') and (y, φ, x) can be composed to $(z, \psi \circ \varphi, x)$ if and only if $y' = y$. Every point x carries an identity element (x, id, x) , and every triple has the inverse (x, φ^{-1}, y) . A set with such a partially defined multiplication, with identities and with inverses is called a *groupoid*, and its elements are called *arrows*. After the restriction to B , the groupoid of triples relates the points of B which occupy the same position in their tiles (such points lie in the same *orbit*) and it still contains the symmetries which fix a given point (its *isotropy group*).

In short, groups describe global homogeneous symmetry, while groupoids generalize them by describing also local and partial symmetries. They reduce to groups when the base consists of one point. In the rest of the chapter we add a smooth structure to this notion and construct its infinitesimal counterpart.

1.2. Lie groupoids

1.2.1. Definitions

Definition 1.1 (Groupoid). A *groupoid* consists of two sets, $\mathcal{G}$ (the *arrows*) and M (the *objects* or *base*), together with structure maps

- (i) *source and target*: $s, t: \mathcal{G} \rightarrow M$;
- (ii) *multiplication*: $m: \mathcal{G}^{(2)} \rightarrow \mathcal{G}$, $(g, h) \mapsto gh$, defined on the set of *composable pairs*
 $\mathcal{G}^{(2)} = \{(g, h) \in \mathcal{G} \times \mathcal{G} : s(g) = t(h)\}$;
- (iii) *unit*: $u: M \rightarrow \mathcal{G}$, $x \mapsto 1_x$;
- (iv) *inverse*: $\iota: \mathcal{G} \rightarrow \mathcal{G}$, $g \mapsto g^{-1}$;

satisfying, whenever the relevant products are defined, the compatibility and associativity identities

$$s(gh) = s(h), \quad t(gh) = t(g), \quad (gh)k = g(hk),$$

and the unit and inverse identities

$$s(1_x) = t(1_x) = x, \quad 1_{t(g)} g = g = g 1_{s(g)}, \quad g g^{-1} = 1_{t(g)}, \quad g^{-1} g = 1_{s(g)}.$$

We denote an arrow $g \in \mathcal{G}$ with $s(g) = x$ and $t(g) = y$ as $y \xleftarrow{g} x$, and a groupoid as $\mathcal{G} \rightrightarrows M$ to record the two maps s, t from arrows to objects. Two arrows compose precisely when the source of the first one matches the target of the second.

Equivalently, a groupoid is a small category in which every arrow is invertible [13, 3]. If $M = \{*\}$ is a single point, then every pair of arrows is composable, and the structure reduces to a group.

A groupoid in the sense of Definition 1.1 is sometimes called *set-theoretic*, to emphasise that no topology or smooth structure is assumed. If $\mathcal{G}$ and M carry topologies and all structure maps are continuous, the groupoid is *topological*.

Definition 1.2 (Lie groupoid). A *Lie groupoid* is a groupoid $\mathcal{G} \rightrightarrows M$ in which $\mathcal{G}$ and M are smooth manifolds, all structure maps are smooth, and the source and target maps $s, t: \mathcal{G} \rightarrow M$ are surjective submersions [1, 3, 2].

The submersion condition on s and t has two consequences. First, the set $\mathcal{G}^{(2)}$ of composable pairs is a smooth embedded submanifold of $\mathcal{G} \times \mathcal{G}$. Second, every source fibre $s^{-1}(x)$ and every target fibre $t^{-1}(x)$ is a closed embedded submanifold of $\mathcal{G}$ [1].

Remark 1.1. Smoothness is a genuine additional requirement. Many groupoids that arise naturally are only topological. For this reason Definition 1.1 is kept alongside Definition 1.2.

1.2.2. First properties and notation

For $x, y \in M$ write

$$\mathcal{G}_x = s^{-1}(x), \quad \mathcal{G}^y = t^{-1}(y), \quad \mathcal{G}_x^y = \mathcal{G}_x \cap \mathcal{G}^y$$

for the *source fibre*, the *target fibre*, and the set of arrows from x to y . The *orbit* of $x \in M$ is

$$\mathcal{O}_x = t(\mathcal{G}_x) = \{y \in M : \mathcal{G}_x^y \neq \emptyset\},$$

the set of objects connected from x by an arrow. The orbits partition M . The *isotropy group* (or *vertex group*) at x is

$$\mathcal{G}_x^x = \{g \in \mathcal{G} : s(g) = t(g) = x\}.$$

It is a group under the restricted multiplication, and for a Lie groupoid it is a Lie group [1]. A Lie groupoid is *transitive* if it has a single orbit.

The unit map $u: M \rightarrow \mathcal{G}$ is a smooth embedding: $s \circ u = \text{id}_M$ shows that u is an injective immersion, and its image $u(M)$, which we identify with M , is a closed embedded submanifold of $\mathcal{G}$ [1].

The analogues of left and right multiplication on a group are the *translations* between fibres. Fix $g \in \mathcal{G}$ with $s(g) = x$ and $t(g) = y$. Left multiplication by g ,

$$L_g: \mathcal{G}^x \longrightarrow \mathcal{G}^y, \quad L_g(h) = gh,$$

is a diffeomorphism between target fibres, with inverse $L_{g^{-1}}$. Right multiplication $R_g: \mathcal{G}_y \rightarrow \mathcal{G}_x$, $R_g(h) = hg$, is a diffeomorphism between source fibres. Left translations are used in Section 1.3 to construct the Lie algebroid of $\mathcal{G}$.

1.2.3. Examples

The following three Lie groupoids are used throughout the thesis.

Example 1.1 (Lie group). A Lie group G is a Lie groupoid $G \rightrightarrows *$ over a point. It has a single orbit, and its isotropy group is G itself.

Example 1.2 (Pair groupoid). For any smooth manifold M , the product $M \times M$ is a Lie groupoid over M , the *pair groupoid*, with

$$s(y, x) = x, \quad t(y, x) = y, \quad (z, y)(y, x) = (z, x), \quad u(x) = (x, x), \quad (y, x)^{-1} = (x, y).$$

Thus (y, x) is the unique arrow from x to y . Source and target are the two projections, hence submersions, and the multiplication is smooth. The pair groupoid is transitive with trivial isotropy groups.

Example 1.3 (Action groupoid). Let a Lie group G act smoothly on a manifold M from the left. The *action groupoid* $G \ltimes M \rightrightarrows M$ has as arrows the pairs $(g, x) \in G \times M$, with

$$s(g, x) = x, \quad t(g, x) = g \cdot x, \quad (h, g \cdot x)(g, x) = (hg, x),$$

unit $u(x) = (e, x)$ and inverse $(g, x)^{-1} = (g^{-1}, g \cdot x)$. The orbits and isotropy groups of $G \ltimes M$ coincide with the orbits and stabilisers of the action. The groupoid of triples in Section 1.1 is the action groupoid of $G_{\mathcal{T}}$ acting on $\mathbb{R}^2$.

The three examples display the two extreme cases and a mixture of them: the Lie group has a trivial base and full isotropy, the pair groupoid is transitive with trivial isotropy, and the action groupoid has, in general, non-trivial orbits and non-trivial isotropy groups.

1.3. From Lie groupoids to Lie algebroids

Every Lie group determines a Lie algebra by linearisation of the multiplication at the identity. This section makes a similar construction for a Lie groupoid. The resulting object is a *Lie algebroid* [1, 3]. We first construct it from a Lie groupoid and then provide the abstract definition.

1.3.1. Construction of the Lie algebroid

Recall first the group case. A vector field $X \in \mathfrak{X}(G)$ on a Lie group G is *left-invariant* if $(L_g)_*X = X$ for every $g \in G$. Since each $(L_g)_*$ preserves the Lie bracket, the left-invariant vector fields $\mathfrak{X}^L(G)$ form a Lie subalgebra of $\mathfrak{X}(G)$. Evaluation at the identity is a linear isomorphism $\mathfrak{X}^L(G) \rightarrow T_e G$ with inverse $v \mapsto \overleftarrow{v}$, $\overleftarrow{v}_g = T_e L_g(v)$. The bracket transported to $T_e G$ defines the Lie algebra $\mathfrak{g} = \text{Lie}(G)$. Two features of this construction generalise: a left-invariant vector field is determined by its values at the identities, and its bracket is inherited from the bracket of vector fields. A Lie groupoid has an identity arrow 1_x at every

point $x \in M$, so the values at the identities form a section of a vector bundle over M rather than a single tangent vector.

Let $\mathcal{G} \rightrightarrows M$ be a Lie groupoid. A vector field $X \in \mathfrak{X}(\mathcal{G})$ is *left-invariant* if

- (i) X is tangent to the target fibres: $T_g t(X_g) = 0$ for all $g \in \mathcal{G}$; and
- (ii) X is equivariant under left translation: $T_h L_g(X_h) = X_{gh}$ for every $(g, h) \in \mathcal{G}^{(2)}$.

Condition (i) says that at every arrow g the vector X_g is tangent to the target fibre $\mathcal{G}^{t(g)}$. Left translations map target fibres to target fibres (Section 1.2), so condition (ii) makes sense, and it determines X on the whole $\mathcal{G}$ from the values at the units.

The set $\mathfrak{X}^L(\mathcal{G})$ of left-invariant vector fields is a Lie subalgebra of $\mathfrak{X}(\mathcal{G})$. Tangency to the target fibres is preserved by the bracket: if $X(f \circ t) = Y(f \circ t) = 0$ for all $f \in C^\infty(M)$, then $[X, Y](f \circ t) = X(Y(f \circ t)) - Y(X(f \circ t)) = 0$. Left-invariance is preserved because each L_g is a diffeomorphism between target fibres, so $(L_g)_*[X, Y] = [(L_g)_*X, (L_g)_*Y] = [X, Y]$.

Define the vector bundle

$$A(\mathcal{G}) := \ker(Tt)|_{u(M)} \longrightarrow M,$$

with fibre $A(\mathcal{G})_x = \ker(T_{1_x} t)$, the tangent vectors at the identity arrow 1_x that are tangent to the target fibre $\mathcal{G}^x$. Such a vector is an infinitesimal deformation of the unit arrow 1_x which keeps the target fixed. The map $\Phi: \Gamma(A(\mathcal{G})) \rightarrow \mathfrak{X}^L(\mathcal{G})$,

$$\Phi(\alpha)_g = \overleftarrow{\alpha}_g := T_{1_{s(g)}} L_g(\alpha_{s(g)}),$$

is an isomorphism with inverse $\Phi^{-1}(X) = X|_{u(M)}$. Transporting the bracket of left-invariant vector fields through Φ defines a Lie bracket on $\Gamma(A(\mathcal{G}))$,

$$[\alpha, \beta] := \Phi^{-1}([\overleftarrow{\alpha}, \overleftarrow{\beta}]).$$

We call $A(\mathcal{G})$ the *Lie algebroid of the Lie groupoid* $\mathcal{G}$.

1.3.2. The anchor and the Leibniz rule

A left-invariant vector field is tangent to the target fibres but, in general, not to the source fibres. Applying Ts produces a vector field on the base. Define the *anchor* $\rho: A(\mathcal{G}) \rightarrow TM$ by

$$\rho(\alpha_x) = T_{1_x} s(\alpha_x) \in T_x M, \quad \alpha_x \in A(\mathcal{G})_x = \ker(T_{1_x} t).$$

The anchor measures the displacement of the source: $\rho(\alpha)$ is the vector field induced on M by the flow of $\overleftarrow{\alpha}$. Elements with $\rho(\alpha_x) = 0$ move neither the source nor the target. They form the *isotropy Lie algebra* $\mathfrak{g}_x = \ker \rho_x$, which is the Lie algebra of the vertex group $\mathcal{G}_x^x$.

The bracket on $\Gamma(A(\mathcal{G}))$ is $\mathbb{R}$ -bilinear but not $C^\infty(M)$ -bilinear: for all $\alpha, \beta \in \Gamma(A(\mathcal{G}))$ and $f \in C^\infty(M)$,

$$[\alpha, f\beta] = f[\alpha, \beta] + \rho(\alpha)(f)\beta. \tag{1.3.1}$$

To prove (1.3.1), we note first that the left-invariant extension of $f\beta$ is $(f \circ s) \overleftarrow{\beta}$, where $f \circ s$ is the pull-back of f to $\mathcal{G}$ along the source map. The Leibniz rule for vector fields on $\mathcal{G}$ gives

$$[\overleftarrow{\alpha}, (f \circ s) \overleftarrow{\beta}] = (f \circ s) [\overleftarrow{\alpha}, \overleftarrow{\beta}] + \overleftarrow{\alpha}(f \circ s) \overleftarrow{\beta}.$$

Now $\overleftarrow{\alpha}$ is s -related to $\rho(\alpha)$, so $\overleftarrow{\alpha}(f \circ s) = \rho(\alpha)(f) \circ s$, and the restriction to the unit section gives (1.3.1). The correction term $\rho(\alpha)(f)\beta$ vanishes when $\rho = 0$ or when f is constant. Because of this term, the operator $[\alpha, \cdot]$ is a first-order differential operator. The value of $[\alpha, \beta]$ at x depends on $\beta(x)$ and also on the derivative of β in the direction $\rho(\alpha)_x$.

1.3.3. The abstract definition

Definition 1.3 (Lie algebroid). A *Lie algebroid* over a manifold M is a vector bundle $A \rightarrow M$ equipped with a Lie bracket $[\cdot, \cdot]_A$ on $\Gamma(A)$ and a vector bundle morphism $\rho: A \rightarrow TM$, called the *anchor*, satisfying the Leibniz rule

$$[\alpha, f\beta]_A = f[\alpha, \beta]_A + \rho(\alpha)(f)\beta \quad \text{for all } \alpha, \beta \in \Gamma(A), f \in C^\infty(M).$$

By construction, $A(\mathcal{G})$ is a Lie algebroid over M . The compatibility of the anchor with the two brackets does not have to be assumed, since it follows from the definition.

Proposition 1.1. *In any Lie algebroid $A \rightarrow M$ the anchor preserves brackets:*

$$\rho([\alpha, \beta]_A) = [\rho(\alpha), \rho(\beta)] \quad \text{for all } \alpha, \beta \in \Gamma(A).$$

Proof. Let $\alpha, \beta, \gamma \in \Gamma(A)$ and $f \in C^\infty(M)$. We expand $[\alpha, [\beta, f\gamma]_A]_A$ in two ways. First we apply the Leibniz rule twice directly. Then we use the Jacobi identity to write the same expression as $[[\alpha, \beta]_A, f\gamma]_A + [\beta, [\alpha, f\gamma]_A]_A$ and we apply the Leibniz rule to each term. We compare the two expansions. The terms in which f multiplies a bracket of sections agree by the Jacobi identity for $[\cdot, \cdot]_A$, and the terms with a single derivative of f agree term by term. The remaining terms contain a second derivative of f and a factor γ , which gives

$$\rho(\alpha)(\rho(\beta)(f)) = \rho([\alpha, \beta]_A)(f) + \rho(\beta)(\rho(\alpha)(f)).$$

Hence $\rho([\alpha, \beta]_A)(f) = [\rho(\alpha), \rho(\beta)](f)$ for every f , which proves the claim. $\square$

Remark 1.2. The anchor is a morphism from $(\Gamma(A), [\cdot, \cdot]_A)$ to $(\mathfrak{X}(M), [\cdot, \cdot])$. Its image $\text{im } \rho \subseteq TM$ is an involutive distribution, perhaps of varying rank, whose integral submanifolds are the *orbits* of the algebroid. Its kernel $\mathfrak{g}_x = \ker \rho_x$ at each point is a Lie algebra, the *isotropy Lie algebra*. For $A = A(\mathcal{G})$ the orbits and the isotropy Lie algebras are the same as for the groupoid $\mathcal{G}$.

Remark 1.3. A Lie algebroid is called *integrable* if it is isomorphic to $A(\mathcal{G})$ for some Lie groupoid $\mathcal{G} \rightrightarrows M$. In contrast to the theory of Lie groups and Lie algebras, not every Lie algebroid is integrable: there is a genuine obstruction [15]. The general theory is not needed in this thesis, integrability becomes meaningful only for the integration of Poisson manifolds, treated in Chapter 3.

1.3.4. Examples

We compute the Lie algebroids of the three groupoids of Section 1.2.3.

Example 1.4 (Lie algebra). A finite-dimensional Lie algebra $\mathfrak{g}$ is a Lie algebroid over a point, with zero anchor and its own bracket. It is the Lie algebroid of any Lie group integrating it (Example 1.1). Since $TM = 0$, the correction term in the Leibniz rule vanishes.

Example 1.5 (Tangent bundle). The tangent bundle $TM \rightarrow M$, with the Lie bracket of vector fields and with the anchor $\rho = \text{id}_{TM}$, is a Lie algebroid. It is the Lie algebroid of the pair groupoid $M \times M \rightrightarrows M$ (Example 1.2). Indeed, for $t(y, x) = y$ the target fibre over x is $\{x\} \times M$, and its tangent space at the unit (x, x) is identified with $T_x M$, so $A(M \times M) \cong TM$. The anchor $\rho = Ts$ is the identity map, because s is the projection onto the second factor. The kernel $\ker \rho = 0$ corresponds to the trivial isotropy of the pair groupoid.

Example 1.6 (Action algebroid). Let a Lie group G act on M from the left, with fundamental vector fields $a: \mathfrak{g} \rightarrow \mathfrak{X}(M)$, $a(\xi)_x = \frac{d}{d\varepsilon}|_0 \exp(\varepsilon\xi) \cdot x$. We compute the Lie algebroid of the action groupoid $G \ltimes M \rightrightarrows M$ of Example 1.3.

The bundle and the anchor. Differentiate $t(g, x) = g \cdot x$ at the unit (e, x) along a curve $\varepsilon \mapsto (\exp(\varepsilon\xi), x_\varepsilon)$ with velocity (ξ, v) . This gives

$$T_{(e,x)}t(\xi, v) = a(\xi)_x + v.$$

So the fibre of the Lie algebroid is

$$A(G \ltimes M)_x = \ker T_{(e,x)}t = \{ (\xi, -a(\xi)_x) : \xi \in \mathfrak{g} \},$$

and the first component identifies it with $\mathfrak{g}$. So the Lie algebroid is the trivial bundle $A = M \times \mathfrak{g}$, called the *action algebroid* $\mathfrak{g} \ltimes M$. Its sections are the $\mathfrak{g}$ -valued functions $\sigma \in C^\infty(M, \mathfrak{g})$. The anchor is Ts , and $s(g, x) = x$ returns the second component. So on constant sections

$$\rho(\xi)_x = -a(\xi)_x.$$

Left-invariant extensions. Left translation by $(g, h \cdot x)$ sends (h, x) to (gh, x) . So its differential sends (w, v) to $(T_h L_g w, v)$. By the construction of Section 1.3.1 the left-invariant extension of the constant section ξ is

$$\overleftarrow{\xi}(g, x) = (T_e L_g \xi, -a(\xi)_x).$$

The two components separate. The first is the left-invariant field ξ^L on G , which does not depend on x . The second is the field $-a(\xi)$ on M , which does not depend on g .

The bracket on constant sections. Each component is constant in the other variable. So the bracket of vector fields can be separated into different components. So

$$[\overleftarrow{\xi}, \overleftarrow{\eta}] = ([\xi^L, \eta^L], [a(\xi), a(\eta)]) = ([\xi, \eta]^L, -a([\xi, \eta])) = \overleftarrow{[\xi, \eta]}.$$

The second component uses $[a(\xi), a(\eta)] = -a([\xi, \eta])$, the reversal of the bracket of a left action. Restricting to the units gives $[\xi, \eta]_A = [\xi, \eta]$ on constant sections. The reversal makes the second components match. It also makes the anchor a homomorphism, since $\rho([\xi, \eta]) = -a([\xi, \eta]) = [a(\xi), a(\eta)] = [\rho(\xi), \rho(\eta)]$, as Proposition 1.1 requires.

The bracket on all sections. Fix a basis $\xi_1, \dots, \xi_n$ of $\mathfrak{g}$ and write $\sigma = \sigma^i \xi_i$ and $\tau = \tau^j \xi_j$ with coefficients in $C^\infty(M)$. The Leibniz rule (1.3.1), its mirror through antisymmetry, and $\sigma^i \rho(\xi_i) = \rho(\sigma)$ give

$$[\sigma, \tau]_A = \sigma^i \tau^j [\xi_i, \xi_j] + \rho(\sigma)(\tau^j) \xi_j - \rho(\tau)(\sigma^i) \xi_i = [\sigma, \tau]_{\mathfrak{g}} - a(\sigma) \tau + a(\tau) \sigma.$$

Here $[\sigma, \tau]_{\mathfrak{g}}$ is the pointwise bracket, and $a(\sigma) \tau$ is the derivative of the $\mathfrak{g}$ -valued function τ along the vector field $x \mapsto a(\sigma(x))_x$. Neither side depends on the choice of basis.

Chapter 2

Poisson Manifolds and Cotangent Lie Algebroids

This chapter develops the Poisson geometry used in the rest of the thesis. It has two deliverables. The first is the *symplectic foliation* of a Poisson manifold, together with its Casimir functions: this is the geometry that the structure-preserving integrators of Chapter 4 are built to respect. The second is the *cotangent Lie algebroid* T^*M , the infinitesimal object that Chapter 3 integrates to a symplectic groupoid. The dual $\mathfrak{g}^*$ of a Lie algebra runs through both as the worked example, and it is where the free rigid body lives.

Section 2.1 defines Poisson manifolds through the properties of the symplectic bracket that survive without non-degeneracy. Section 2.2 treats Hamiltonian dynamics, Casimir functions and the decomposition into symplectic leaves. Section 2.3 studies the Lie–Poisson structure on $\mathfrak{g}^*$, its coadjoint orbits and the free rigid body. Section 2.4 constructs the cotangent Lie algebroid.

The material is standard and is treated in the books of Vaisman [7], Crainic, Fernandes and Mărcuț [6], Cannas da Silva and Weinstein [4] and, for the mechanics, Marsden and Ratiu [8]. More precise references are given at each step.

2.1. From symplectic to Poisson manifolds

The simplest models of conservative mechanical systems have symplectic phase spaces. Their defining 2-form is non-degenerate. This is a strong assumption which fails after reduction by a symmetry, and under constraints. What survives in these situations is the Poisson bracket on the algebra of smooth functions. This section recalls the symplectic bracket, then uses the properties that do not use non-degeneracy to define a Poisson manifold.

Let (M, ω) be a symplectic manifold. Non-degeneracy means that the bundle map

$$\omega^\flat: TM \longrightarrow T^*M, \quad v \longmapsto \iota_v \omega,$$

is an isomorphism. Each $H \in C^\infty(M)$ then has a unique *Hamiltonian vector field* X_H defined by

$$\iota_{X_H} \omega = dH, \quad \text{that is} \quad X_H = (\omega^\flat)^{-1}(dH).$$

The trajectories of the system are its integral curves. The *Poisson bracket* of two functions is

$$\{f, g\} = \omega(X_f, X_g),$$

and a short computation gives $X_H(f) = \{f, H\}$. So every observable evolves along the flow by $\dot{f} = \{f, H\}$.

This bracket has four properties. It is bilinear and antisymmetric. It satisfies the *Leibniz rule*

$$\{f, gh\} = \{f, g\}h + g\{f, h\},$$

because a Hamiltonian vector field is a derivation of $C^\infty(M)$. And it satisfies the *Jacobi identity*

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0,$$

which is the only one that uses $d\omega = 0$. Keeping the four properties without mentioning ω gives the general notion.

Definition 2.1 (Poisson manifold). A *Poisson manifold* is a smooth manifold M with a bilinear map

$$\{\cdot, \cdot\}: C^\infty(M) \times C^\infty(M) \longrightarrow C^\infty(M)$$

that is antisymmetric, satisfies the Jacobi identity, and is a derivation in each argument,

$$\{f, gh\} = \{f, g\}h + g\{f, h\} \quad \text{for all } f, g, h \in C^\infty(M).$$

The map $\{\cdot, \cdot\}$ is called the *Poisson bracket* [6, 7].

The first two axioms make $C^\infty(M)$ a Lie algebra. As in the symplectic case, each $H \in C^\infty(M)$ has a *Hamiltonian vector field*

$$X_H = \{\cdot, H\}.$$

The Leibniz rule makes $\{\cdot, H\}$ a derivation of $C^\infty(M)$, observables again evolve by $\dot{f} = \{f, H\}$. The difference from the symplectic case is that X_H does not always determine H , and not every vector field is Hamiltonian.

We now record a bivector form of the bracket.

Remark 2.1. A map $C^\infty(M)^k \rightarrow C^\infty(M)$ that is $\mathbb{R}$ -multilinear and a derivation in each argument (a *k-derivation*) has, in any chart,

$$D(f_1, \dots, f_k) = \sum_{i_1, \dots, i_k} D(x^{i_1}, \dots, x^{i_k}) \partial_{i_1} f_1 \cdots \partial_{i_k} f_k.$$

So its value at x depends only on the differentials there, and $D \equiv 0$ as soon as it vanishes on coordinate functions. An antisymmetric 2-derivation is therefore $\{f, g\} = \pi(df, dg)$ for a unique bivector field $\pi \in \Gamma(\wedge^2 TM)$ with $\pi^{ij} = \{x^i, x^j\}$ [6].

By Remark 2.1 the value $\{f, g\}(x)$ depends only on df_x and dg_x . It is an antisymmetric 2-derivation, so there is a unique bivector field $\pi \in \Gamma(\wedge^2 TM)$, the *Poisson bivector*, with

$$\{f, g\} = \pi(df, dg) \quad \text{for all } f, g \in C^\infty(M).$$

The bivector defines a bundle map

$$\pi^\sharp: T^*M \longrightarrow TM, \quad \langle \alpha, \pi^\sharp \beta \rangle = \pi(\alpha, \beta),$$

the Poisson analogue of $(\omega^\flat)^{-1}$. The Hamiltonian vector field is

$$X_H = \pi^\sharp(dH),$$

because $\langle df, \pi^\sharp dH \rangle = \pi(df, dH) = \{f, H\} = X_H(f)$.

Of the four axioms, bilinearity, antisymmetry and the Leibniz rule hold for any bivector field. Only the Jacobi identity is a real condition on π . That condition is

$$[\pi, \pi]_{\text{SN}} = 0,$$

where $[\cdot, \cdot]_{\text{SN}}$ is the Schouten–Nijenhuis bracket, the extension of the Lie bracket of vector fields to multivector fields [7]. We will not need this identity and state it only to record the conclusion: *a Poisson structure on M is the same datum as a bivector field π with $[\pi, \pi]_{\text{SN}} = 0$, the bracket being recovered by $\{f, g\} = \pi(df, dg)$. From now on we write (M, π) for a Poisson manifold and use the bracket and the bivector interchangeably.*

Example 2.1 (Symplectic manifolds). Every symplectic manifold (M, ω) is a Poisson manifold. The bracket $\{f, g\} = \omega(X_f, X_g)$ is antisymmetric and Leibniz by construction, and satisfies the Jacobi identity because $d\omega = 0$. Comparing $X_H = (\omega^\flat)^{-1}(dH)$ with $X_H = \pi^\sharp(dH)$ identifies its bivector as the inverse of the form,

$$\pi^\sharp = (\omega^\flat)^{-1},$$

so here $\pi^\sharp$ is an isomorphism at every point. By Darboux's theorem [5] every symplectic manifold looks, near any point, like $\mathbb{R}^{2n}$ with coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$ in which

$$\{q^i, p_j\} = \delta_j^i, \quad \{q^i, q^j\} = \{p_i, p_j\} = 0,$$

that is $\pi = \sum_i \partial_{q^i} \wedge \partial_{p_i}$. The Hamiltonian vector field of H reproduces Hamilton's equations,

$$\dot{q}^i = \{q^i, H\} = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = \{p_i, H\} = -\frac{\partial H}{\partial q^i}.$$

Remark 2.2 (Rank). The *rank* of a Poisson manifold at x is the rank of the linear map $\pi_x^\sharp: T_x^*M \rightarrow T_xM$, or equivalently of the antisymmetric matrix $(\pi^{ij}(x))$ of components of π in a chart. Being the rank of an antisymmetric map, it is always even. The symplectic Poisson structures are exactly those of maximal rank $\dim M$ at every point, which forces $\dim M$ to be even. The zero bracket has rank 0 everywhere. In general the rank lies between these extremes and may change from point to point.

2.2. Casimirs and the symplectic foliation

The degeneracy of a Poisson bivector has two consequences that a symplectic structure does not show. Both follow from the Jacobi identity. Some functions are conserved by *every* Hamiltonian. They come from the kernel of $\pi^\sharp$ and are called the Casimir functions. The manifold decomposes into symplectic submanifolds, the *leaves*, on which the trajectories stay. They come from the image of $\pi^\sharp$. We treat both only as far as the later chapters need.

2.2.1. Hamiltonian flows and Poisson maps

On a Poisson manifold every $H \in C^\infty(M)$ generates the flow of $X_H = \pi^\sharp(dH)$, along which $\dot{f} = \{f, H\}$. Antisymmetry gives conservation of energy, $\dot{H} = \{H, H\} = 0$. More generally f is a constant of the motion for H when $\{f, H\} = 0$. The Jacobi identity adds two facts:

Proposition 2.1. *On a Poisson manifold, for all $f, g, H \in C^\infty(M)$,*

$$(i) \quad X_H\{f, g\} = \{X_H f, g\} + \{f, X_H g\};$$

$$(ii) \quad [X_f, X_g] = -X_{\{f, g\}}.$$

Both are restatements of the Jacobi identity.

Proof. Write the Jacobi identity in the form $\{a, \{b, c\}\} = \{\{a, b\}, c\} + \{b, \{a, c\}\}$. For (i), take $a = H$, $b = f$, $c = g$ and use $X_H \phi = \{\phi, H\}$ together with antisymmetry:

$$X_H\{f, g\} = \{\{f, g\}, H\} = -\{H, \{f, g\}\} = -\{\{H, f\}, g\} - \{f, \{H, g\}\} = \{X_H f, g\} + \{f, X_H g\}.$$

For (ii), apply both sides to an arbitrary h . Since $X_\phi \psi = \{\psi, \phi\}$,

$$[X_f, X_g]h = \{\{h, g\}, f\} - \{\{h, f\}, g\} = -\{h, \{f, g\}\} = -X_{\{f, g\}}h,$$

where the middle equality is the Jacobi identity. □

Part (i) says that X_H preserves the Poisson bracket. In tensor form this is $\mathcal{L}_{X_H}\pi = 0$. That is because the Schouten formula for the Lie derivative of a bivector [6] gives, for any vector field X ,

$$(\mathcal{L}_X\pi)(df, dg) = X\{f, g\} - \{Xf, g\} - \{f, Xg\},$$

which for $X = X_H$ vanishes by Proposition 2.1(i). A tensor whose Lie derivative along X_H vanishes is invariant under the flow, so the flow ϕ_t of X_H satisfies $(\phi_t)_*\pi = \pi$ wherever it is defined. Maps that preserve the bracket have a name.

Definition 2.2 (Poisson map). A smooth map $\Phi: (M, \pi_M) \rightarrow (N, \pi_N)$ between Poisson manifolds is a *Poisson map* if

$$\{\varphi \circ \Phi, \psi \circ \Phi\}_M = \{\varphi, \psi\}_N \circ \Phi \quad \text{for all } \varphi, \psi \in C^\infty(N).$$

Preservation of π is exactly this condition, so every Hamiltonian flow is a Poisson diffeomorphism. Poisson maps are the natural morphisms between Poisson manifolds.

2.2.2. Casimir functions

The kernel of $\pi^\sharp$ carries the functions that are constant for any dynamics.

Definition 2.3 (Casimir function). A *Casimir function* of (M, π) is a function $C \in C^\infty(M)$ with $\{C, f\} = 0$ for all $f \in C^\infty(M)$.

Since $\{C, f\} = \langle df, \pi^\sharp dC \rangle$, a function C is a Casimir exactly when $\pi^\sharp dC = 0$. That is $X_C = 0$, or equivalently $dC \in \ker \pi^\sharp$ at every point. The dynamical meaning is that for *every* Hamiltonian H ,

$$\dot{C} = \{C, H\} = 0.$$

A Casimir is therefore conserved by every Hamiltonian system on (M, π) at once. On a symplectic manifold $\pi^\sharp$ is injective, so the only Casimirs are the constants. A non-constant Casimir is a sign of degeneracy.

2.2.3. The symplectic foliation

The image of $\pi^\sharp$ carries the directions available to the dynamics. At each point x , as f ranges over $C^\infty(M)$, the vectors $X_f(x) = \pi^\sharp(df_x)$ fill the subspace

$$\mathcal{D}_x = \text{im } \pi_x^\sharp \subseteq T_x M.$$

The assignment $x \mapsto \mathcal{D}_x$ is the *characteristic distribution*. Its dimension is the rank of π , which may change from point to point. So the Frobenius theorem does not apply.

The characteristic distribution $\mathcal{D} = \text{im } \pi^\sharp$ is generated by the family of Hamiltonian vector fields $F = \{X_f : f \in C^\infty(M)\}$: at each x , $\mathcal{D}_x = \{X_f(x) : f \in C^\infty(M)\}$, since every covector is a differential. This family is closed under the Lie bracket, by Proposition 2.1(ii), and $\mathcal{D}$ is invariant under the flows of its members, by Proposition 2.1(i). For a family of vector fields with these properties the generated distribution has non-constant rank in general, so the Frobenius theorem does not apply; integrability is instead guaranteed by the Stefan–Sussmann theorem [6, 7].

Theorem 2.1 (Symplectic foliation). *Let (M, π) be a Poisson manifold.*

- (i) *The characteristic distribution integrates to a foliation of M by immersed submanifolds, the symplectic leaves; the leaf through x has tangent space $\mathcal{D}_x$, and two points lie on the same leaf exactly when a chain of Hamiltonian flows joins them.*

(ii) Each leaf L carries a unique symplectic form, given by

$$\omega_L(\pi^\sharp\alpha, \pi^\sharp\beta) = \pi(\alpha, \beta),$$

for which the inclusion $L \hookrightarrow M$ is a Poisson map.

We take part (i) from [6, 7], as we do the closedness of ω_L in part (ii), which is where the Jacobi identity of π in the Schouten bracket form is used. That ω_L is well defined and non-degenerate is pure linear algebra: $\pi^\sharp\alpha = \pi^\sharp\alpha'$ gives $\pi(\alpha - \alpha', \beta) = -\langle\beta, \pi^\sharp(\alpha - \alpha')\rangle = 0$, non-degenerate because $\omega_L(\pi^\sharp\alpha, \cdot) = 0$ forces $\pi(\alpha, \cdot) = 0$ and hence $\pi^\sharp\alpha = 0$. Since $T_x L = \mathcal{D}_x = \text{im } \pi^\sharp_x$ by part (i), the dimension of L equals the rank of π on L . It is even because $\pi^\sharp$ is antisymmetric.

Since X_H takes values in $\mathcal{D}$, every trajectory of H stays on the leaf through its initial point. On that leaf it is the symplectic flow of the restriction $H|_L$. A Casimir C is constant on each leaf: for $v = \pi^\sharp\alpha \in \mathcal{D}_x$, $dC(v) = \langle dC, \pi^\sharp\alpha \rangle = -\langle \alpha, \pi^\sharp dC \rangle = 0$, since $\pi^\sharp dC = 0$. A symplectic manifold is the case of a single leaf.

Example 2.2 (see [6]). On $\mathbb{R}^3$ with coordinates (x, y, z) take $\pi = \partial_x \wedge \partial_y$, so that

$$\{x, y\} = 1, \quad \{x, z\} = \{y, z\} = 0.$$

The rank is 2 everywhere. The characteristic distribution is spanned by ∂_x and ∂_y , and z is a Casimir, since dz spans $\ker \pi^\sharp$ at every point. No Hamiltonian moves z , so every trajectory stays in a plane $z = \text{const}$, on which π restricts to the standard symplectic structure of $\mathbb{R}^2$. The symplectic leaves are these planes, which are the level sets of the single Casimir z .

2.3. The Lie–Poisson structure and the rigid body

Sections 2.1–2.2 were general theory. We now study one specific Poisson manifold, the dual $\mathfrak{g}^*$ of a Lie algebra. It is the phase space of a mechanical system reduced by a symmetry group, the free rigid body is the main case. Its cotangent algebroid, defined in Section 2.4, is integrated by the groupoid $T^*G \rightrightarrows \mathfrak{g}^*$ used in the latter chapters.

2.3.1. The Lie–Poisson bracket

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra with bracket $[\cdot, \cdot]$, and $\mathfrak{g}^*$ its dual. Because $\mathfrak{g}^*$ is a vector space, its tangent space at any point is canonically again $\mathfrak{g}^*$, and its cotangent space is the double dual,

$$T_\mu \mathfrak{g}^* \cong \mathfrak{g}^*, \quad T_\mu^* \mathfrak{g}^* \cong (\mathfrak{g}^*)^* \cong \mathfrak{g}.$$

So the differential of a function $f \in C^\infty(\mathfrak{g}^*)$ at a point μ is an element of $\mathfrak{g}$,

$$d_\mu f \in \mathfrak{g}, \quad \langle \nu, d_\mu f \rangle = \left. \frac{d}{dt} \right|_{t=0} f(\mu + t\nu) \quad (\nu \in \mathfrak{g}^*),$$

where $\langle \cdot, \cdot \rangle$ is the pairing between $\mathfrak{g}^*$ and $\mathfrak{g}$. For a linear function $\ell_\xi(\mu) = \langle \mu, \xi \rangle$ with $\xi \in \mathfrak{g}$, the differential is constant, $d_\mu \ell_\xi = \xi$. This identification allows us to define the Poisson structure on $\mathfrak{g}^*$ based on the Lie algebra structure of $\mathfrak{g}$.

Definition 2.4 (Lie–Poisson bracket). The *Lie–Poisson bracket* on $\mathfrak{g}^*$ is

$$\{f, g\}(\mu) = \langle \mu, [d_\mu f, d_\mu g] \rangle, \quad f, g \in C^\infty(\mathfrak{g}^*).$$

The bracket is bilinear and antisymmetric. It is a derivation in each argument, since $d_\mu(fg) = f(\mu) d_\mu g + g(\mu) d_\mu f$ and the pairing is linear. On linear functions it returns the Lie bracket,

$$\{\ell_\xi, \ell_\eta\} = \ell_{[\xi, \eta]},$$

because $d\ell_\xi = \xi$ and $d\ell_\eta = \eta$ are constant. Fixing a basis $\{e_i\}$ of $\mathfrak{g}$ with $[e_i, e_j] = c_{ij}^k e_k$ and writing $\mu_i = \ell_{e_i}$ for the resulting linear coordinates on $\mathfrak{g}^*$,

$$\{\mu_i, \mu_j\} = c_{ij}^k \mu_k, \quad \text{so} \quad \{f, g\}(\mu) = c_{ij}^k \mu_k \frac{\partial f}{\partial \mu_i} \frac{\partial g}{\partial \mu_j}.$$

The bracket is thus linear in μ and vanishes at the origin.

The only thing to check is the Jacobi identity. The Jacobiator

$$J(f, g, h) = \{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\}$$

is a derivation in each argument. By Remark 2.1 it vanishes identically as soon as it vanishes on the coordinates μ_i . There,

$$J(\mu_i, \mu_j, \mu_k) = (c_{jk}^m c_{im}^\ell + c_{ki}^m c_{jm}^\ell + c_{ij}^m c_{km}^\ell) \mu_\ell.$$

The coefficient is exactly the Jacobi identity for the structure constants of $\mathfrak{g}$, hence zero. So the Lie–Poisson bracket satisfies the Jacobi identity, and $(\mathfrak{g}^*, \{\cdot, \cdot\})$ is a Poisson manifold.

2.3.2. The equations of motion

On $\mathfrak{g}^*$ the Hamiltonian equations $\dot{f} = \{f, H\}$ take a form written entirely inside the Lie algebra. For $\xi \in \mathfrak{g}$ recall the *adjoint* operator $\text{ad}_\xi \eta = [\xi, \eta]$ and define the *coadjoint* operator $\text{ad}_\xi^*: \mathfrak{g}^* \rightarrow \mathfrak{g}^*$ by

$$\langle \text{ad}_\xi^* \mu, \eta \rangle = \langle \mu, [\xi, \eta] \rangle, \quad \eta \in \mathfrak{g}.$$

Testing $\dot{\mu}$ against a constant $\xi = d\ell_\xi$ and using antisymmetry of the Lie bracket,

$$\langle \dot{\mu}, \xi \rangle = \left. \frac{d}{dt} \right|_{t=0} \ell_\xi(\mu) = \{\ell_\xi, H\}(\mu) = \langle \mu, [\xi, d_\mu H] \rangle = -\langle \mu, [d_\mu H, \xi] \rangle = -\langle \text{ad}_{d_\mu H}^* \mu, \xi \rangle.$$

As ξ is arbitrary, this is the *Lie–Poisson equation*

$$\dot{\mu} = -\text{ad}_{d_\mu H}^* \mu. \quad (2.3.1)$$

The entire dynamics is defined using the coadjoint operator and the single element $d_\mu H \in \mathfrak{g}$.

2.3.3. Coadjoint orbits and the symplectic leaves

The group G acts on its Lie algebra by the adjoint representation Ad_g , and dually on $\mathfrak{g}^*$ by the *coadjoint action*

$$\langle \text{Ad}_g^* \mu, \xi \rangle = \langle \mu, \text{Ad}_{g^{-1}} \xi \rangle.$$

Its orbits $\mathcal{O}_\mu = \{\text{Ad}_g^* \mu : g \in G\}$ are called the *coadjoint orbits*. Differentiating along $\exp(t\xi)$ gives the vector fields generating the action,

$$\left. \frac{d}{dt} \right|_{t=0} \text{Ad}_{\exp(t\xi)}^* \mu = -\text{ad}_\xi^* \mu.$$

The tangent space to the orbit through μ is $T_\mu \mathcal{O}_\mu = \{\text{ad}_\xi^* \mu : \xi \in \mathfrak{g}\}$.

This is exactly the characteristic distribution. The cotangent space $T_\mu^* \mathfrak{g}^*$ is $\mathfrak{g}$, spanned by the differentials $d\ell_\xi = \xi$. So $\text{im } \pi_\mu^\sharp$ is spanned by the Hamiltonian vector fields of the linear functions. By (2.3.1) with $H = \ell_\xi$ the field X_{ℓ_ξ} takes the value $-\text{ad}_\xi^* \mu$ at μ . Hence

$$\text{im } \pi_\mu^\sharp = \{\text{ad}_\xi^* \mu : \xi \in \mathfrak{g}\} = T_\mu \mathcal{O}_\mu.$$

The distribution matches the tangent spaces of the orbits. That the leaves are the connected components of the coadjoint orbits is the theorem of Kirillov, Kostant and Souriau. We take it from [8, 6].

Theorem 2.2 (Kirillov–Kostant–Souriau). *The symplectic leaves of $\mathfrak{g}^*$ are the connected components of the coadjoint orbits. On the orbit through μ the leaf symplectic form is*

$$\omega_\mu(\text{ad}_\xi^* \mu, \text{ad}_\eta^* \mu) = \langle \mu, [\xi, \eta] \rangle, \quad \xi, \eta \in \mathfrak{g}.$$

The formula is the one produced by Theorem 2.1(ii): evaluating $\omega_L(\pi^\sharp d\ell_\xi, \pi^\sharp d\ell_\eta) = \pi(d\ell_\xi, d\ell_\eta)$ gives $\{\ell_\xi, \ell_\eta\}(\mu) = \langle \mu, [\xi, \eta] \rangle$, and the two signs from $\pi^\sharp d\ell_\xi = -\text{ad}_\xi^* \mu$ cancel.

A Casimir is constant on each leaf, hence on each connected component of a coadjoint orbit. So it is invariant under the identity component G_0 . When G is connected, as for $G = \text{SO}(3)$, the components are the full orbits and the Casimirs are exactly the Ad^* -invariant functions.

2.3.4. The free rigid body

The rotating rigid body is the case $G = \text{SO}(3)$. Its configuration is the rotation $R \in \text{SO}(3)$ that carries the body frame to the spatial frame. The spatial frame is fixed in the laboratory. The body frame is attached to the body and turns with it. The choice of frame gives two descriptions of the same motion. In the spatial frame the angular momentum is conserved. In the body frame it is $\mu = I\Omega$ with angular velocity Ω and constant inertia tensor I . The body frame is convenient because I is constant there. The phase space is $T^*\text{SO}(3)$ of dimension six. The energy is kinetic only and depends on Ω and not on the orientation. It is thus invariant under the rotational symmetry. Reduction by that symmetry leaves three coordinates, the body angular momentum $\mu \in \mathfrak{so}(3)^*$. The reduction is carried out in Chapter 4. Here we read the dynamics on $\mathfrak{so}(3)^*$ directly.

We identify both $\mathfrak{so}(3)$ and $\mathfrak{so}(3)^*$ with $\mathbb{R}^3$. This way the bracket, the adjoint and coadjoint operators and the Lie–Poisson bracket all become vector operations. The first identification is the hat map [8]

$$\hat{\cdot}: \mathbb{R}^3 \rightarrow \mathfrak{so}(3), \quad \hat{\xi} \eta = \xi \times \eta.$$

It sends a vector ξ to the matrix that acts as the cross product with ξ . It is a linear isomorphism which respects the brackets. For any ζ

$$[\hat{\xi}, \hat{\eta}] \zeta = \xi \times (\eta \times \zeta) - \eta \times (\xi \times \zeta) = (\xi \times \eta) \times \zeta = \widehat{\xi \times \eta} \zeta,$$

where the middle step is the Jacobi identity of the cross product. So $[\hat{\xi}, \hat{\eta}] = \widehat{\xi \times \eta}$ and the hat map is a Lie algebra isomorphism from $(\mathbb{R}^3, \times)$ onto $\mathfrak{so}(3)$. The adjoint operator is then the cross product,

$$\text{ad}_\xi \eta = [\xi, \eta] = \xi \times \eta,$$

by the bracket identity just proved.

Rotations preserve the cross product, $R(a \times b) = (Ra) \times (Rb)$ for $R \in \text{SO}(3)$. Acting on a vector v we get $R\hat{\xi}R^{-1}v = R(\xi \times R^{-1}v) = (R\xi) \times v$. So

$$\text{Ad}_R \hat{\xi} = \widehat{R\xi}.$$

Conjugation on $\mathfrak{so}(3)$ is rotation of the underlying vector.

The second identification uses the dot product. It is a nondegenerate pairing, so every functional on $\mathfrak{so}(3) \cong \mathbb{R}^3$ is $\eta \mapsto m \cdot \eta$ for a unique $m \in \mathbb{R}^3$. This identifies $\mathfrak{so}(3)^*$ with $\mathbb{R}^3$,

$$\langle \mu, \hat{\eta} \rangle = m \cdot \eta, \quad \mu \leftrightarrow m.$$

The dot product is also $\text{SO}(3)$ -invariant since R is orthogonal. This is what turns the coadjoint action into a rotation:

$$\langle \text{Ad}_R^* \mu, \hat{\xi} \rangle = \langle \mu, \text{Ad}_{R^{-1}} \hat{\xi} \rangle = \langle \mu, \widehat{R^{-1}\xi} \rangle = m \cdot (R^{-1}\xi) = (Rm) \cdot \xi,$$

using $R^{-1} = R^\top$. So $\text{Ad}_R^* \mu = R\mu$. The coadjoint action is the plain rotation and its orbits are the spheres $|\mu| = \text{const}$ and the origin.

The coadjoint operator follows. By its definition

$$\langle \text{ad}_\xi^* \mu, \eta \rangle = \langle \mu, [\xi, \eta] \rangle = \mu \cdot (\xi \times \eta) = (\mu \times \xi) \cdot \eta,$$

where the last step is the scalar triple product identity. This holds for every η so

$$\text{ad}_\xi^* \mu = \mu \times \xi.$$

It remains to find the bracket. A function f on $\mathfrak{so}(3)^*$ has a differential $d_\mu f \in \mathfrak{g}$ by Section 2.3.1. The pairing is now the dot product so

$$\nu \cdot d_\mu f = \left. \frac{d}{dt} \right|_{t=0} f(\mu + t\nu) = \nu \cdot \nabla f,$$

hence $d_\mu f = \nabla f$ is the ordinary gradient. Placing this into Definition 2.4 gives the Lie–Poisson bracket in vector form

$$\{f, g\}(\mu) = \langle \mu, [d_\mu f, d_\mu g] \rangle = \mu \cdot (\nabla f \times \nabla g).$$

So the bracket on $\mathfrak{so}(3)^*$ is not new. It is Definition 2.4 written with vectors.

Remark 2.3 (Body frame and the sign). The bracket of the rigid body is not Definition 2.4 itself. The body lives on $T^*\text{SO}(3)$ with the canonical bracket. Its momentum μ is the canonical momentum read in the body frame, carried to the identity by left translation. Reducing the canonical bracket in this variable gives the *minus* Lie–Poisson bracket

$$\{f, g\}(\mu) = -\langle \mu, [d_\mu f, d_\mu g] \rangle.$$

The spatial frame would give the plus bracket instead. Both share leaves, Casimirs and orbits, but the equation of motion keeps the sign, here $\dot{\mu} = +\text{ad}_{d_\mu H}^* \mu$. We adopt the minus bracket and derive the reduction in Chapter 4 following [8].

On $\mathfrak{so}(3)^* \cong \mathbb{R}^3$ the minus bracket is

$$\{f, g\}(\mu) = -\mu \cdot (\nabla f \times \nabla g).$$

The function $C(\mu) = \frac{1}{2}|\mu|^2$ is a Casimir. Indeed $\nabla C = \mu$ and $\{C, f\} = -\mu \cdot (\mu \times \nabla f) = 0$. Its level sets are the spheres, as Theorem 2.2 requires. A body with principal moments of inertia I_1, I_2, I_3 has energy

$$H(\mu) = \frac{1}{2} \mu \cdot I^{-1} \mu, \quad I = \text{diag}(I_1, I_2, I_3),$$

and $\nabla H = I^{-1} \mu = \Omega$ is the angular velocity. The equation of motion $\dot{\mu} = \text{ad}_{\nabla H}^* \mu = \mu \times \nabla H$ becomes Euler’s equation

$$\dot{\mu} = \mu \times \Omega. \tag{2.3.2}$$

With $\mu = I\Omega$ the first component is $I_1 \dot{\Omega}_1 = (I_2 - I_3) \Omega_2 \Omega_3$, and the cyclic rotation $1 \rightarrow 2 \rightarrow 3$ gives the other two. This is the classical Euler equation and it confirms the sign of the remark.

2.4. The cotangent Lie algebroid

Chapter 1 built the Lie algebroid of a Lie groupoid by differentiation at the units. Chapter 3 will integrate a Poisson manifold by a groupoid. In order to do that, this section builds an algebroid on the cotangent bundle T^*M of a Poisson manifold (M, π) .

2.4.1. Anchor and bracket

We look for a Lie algebroid structure on $A = T^*M$. By Definition 1.3 it is an anchor $\rho: T^*M \rightarrow TM$ and a Lie bracket on $\Omega^1(M)$ with the Leibniz rule. Two requirements fix both.

The anchor is fixed by the dynamics. We demand that the algebroid generates the Hamiltonian flows, so we require $\rho(dH) = X_H = \pi^\sharp(dH)$ for every H . At a point x every covector is $d_x H$ for some H , and ρ_x is linear, so this forces $\rho = \pi^\sharp$.

The bracket is fixed on exact forms. On the differentials it must return the Poisson bracket of the functions. If $(T^*M, \pi^\sharp, [\cdot, \cdot])$ is an algebroid then its anchor preserves brackets (Proposition 1.1), so

$$\pi^\sharp[df, dg] = [\pi^\sharp df, \pi^\sharp dg] = [X_f, X_g] = -X_{\{f, g\}} = \pi^\sharp(-d\{f, g\}),$$

by Proposition 2.1(ii). This tells us to set

$$[df, dg]_\pi = -d\{f, g\}. \quad (2.4.1)$$

The Leibniz rule then carries the bracket from exact forms to all of $\Omega^1(M)$, because locally every one-form is a combination $\sum_i f_i dg_i$. Doing this gives a closed formula, which we take as the definition.

Definition 2.5 (Koszul bracket). Let (M, π) be a Poisson manifold. The *Koszul bracket* on $\Omega^1(M)$ is

$$[\alpha, \beta]_\pi = \mathcal{L}_{\pi^\sharp \alpha} \beta - \mathcal{L}_{\pi^\sharp \beta} \alpha + d(\pi(\alpha, \beta)).$$

On exact forms it reduces to (2.4.1). Using $\pi^\sharp df = X_f$ with $X_f(h) = \{h, f\}$ and $\mathcal{L}_X dh = d(Xh)$,

$$[df, dg]_\pi = d\{g, f\} - d\{f, g\} + d\{f, g\} = -d\{f, g\}.$$

For any bivector π the formula is antisymmetric. It also obeys the Leibniz rule with anchor $\pi^\sharp$. With $\mathcal{L}_{hX} \alpha = h\mathcal{L}_X \alpha + \alpha(X)dh$ and $\pi^\sharp(h\beta) = h\pi^\sharp \beta$ we get

$$[\alpha, h\beta]_\pi = h[\alpha, \beta]_\pi + (\pi^\sharp \alpha \cdot h)\beta + (\pi(\alpha, \beta) - \alpha(\pi^\sharp \beta))dh.$$

The last term is zero because $\alpha(\pi^\sharp \beta) = \pi(\alpha, \beta)$ is the definition of $\pi^\sharp$. So antisymmetry and the Leibniz rule hold for every bivector. Only the Jacobi identity can fail. In Section 2.1 the single condition on π was the Jacobi identity of the bracket, that is $[\pi, \pi]_{\text{SN}} = 0$. The Jacobi identity of the Koszul bracket is the same condition.

Theorem 2.3. Let (M, π) be a Poisson manifold. Then T^*M , with anchor $\pi^\sharp$ and the Koszul bracket, is a Lie algebroid. It is the unique Lie algebroid structure on T^*M with anchor $\pi^\sharp$ that restricts to (2.4.1) on exact forms. We call it the cotangent Lie algebroid of (M, π) .

Proof. Antisymmetry and the Leibniz rule hold for any bivector. We showed this above. Only the Jacobi identity needs π to be Poisson. We prove it in three steps.

First, we check that the anchor respects the bracket. We claim that

$$\pi^\sharp[\alpha, \beta]_\pi = [\pi^\sharp \alpha, \pi^\sharp \beta] \quad (2.4.2)$$

for all one-forms α and β . Both sides are antisymmetric. Under $\beta \mapsto h\beta$ each side gains the term $(\pi^\sharp \alpha \cdot h)\pi^\sharp \beta$ by the Leibniz rule of the corresponding bracket. So the difference of the two sides is $C^\infty(M)$ -bilinear. Then it is enough to check (2.4.2) on exact forms. There it becomes $[X_f, X_g] = -X_{\{f, g\}}$, which is Proposition 2.1(ii).

Second, we check that the Jacobiator

$$J(\alpha, \beta, \gamma) = [[\alpha, \beta]_\pi, \gamma]_\pi + [[\beta, \gamma]_\pi, \alpha]_\pi + [[\gamma, \alpha]_\pi, \beta]_\pi$$

is a tensor. It is antisymmetric, so it is enough to test the last argument. Replace γ by $h\gamma$ and expand by the Leibniz rule. The terms carrying no derivative of h add up to $hJ(\alpha, \beta, \gamma)$. The terms carrying one derivative of h cancel in pairs by antisymmetry. The terms carrying two derivatives of h and a factor γ add up to

$$(\pi^\sharp[\alpha, \beta]_\pi - [\pi^\sharp \alpha, \pi^\sharp \beta])(h)\gamma.$$

This vanishes because the anchor respects the bracket. So $J(\alpha, \beta, h\gamma) = hJ(\alpha, \beta, \gamma)$. Similar computation appears in Proposition 1.1. Here we make it for the Koszul bracket.

A tensor is fixed by its values on exact forms. They span $\Omega^1(M)$ over $C^\infty(M)$. On exact forms using (2.4.1) twice we get

$$J(df, dg, dh) = d(\{\{f, g\}, h\} + \{\{g, h\}, f\} + \{\{h, f\}, g\}) = 0$$

by the Jacobi identity for $\{\cdot, \cdot\}$. So $J \equiv 0$. The Koszul bracket is a Lie bracket and $(T^*M, \pi^\sharp, [\cdot, \cdot]_\pi)$ is a Lie algebroid.

For uniqueness, two algebroid brackets with anchor $\pi^\sharp$ differ by a tensor. Equation (2.4.1) fixes that tensor on the exact forms, which span. $\square$

2.4.2. Examples

Example 2.3 (Symplectic manifolds). Let (M, ω) be symplectic, so $\pi^\sharp = (\omega^\flat)^{-1}$ is an isomorphism (Example 2.1). It is an isomorphism of Lie algebroids between T^*M and the tangent algebroid $(TM, [\cdot, \cdot], \text{id})$, which is the algebroid of the pair groupoid (Examples 1.2 and 1.5). The anchors agree because $\text{id} \circ \pi^\sharp = \pi^\sharp$. On exact forms the brackets agree, since

$$\pi^\sharp[df, dg]_\pi = \pi^\sharp(-d\{f, g\}) = -X_{\{f, g\}} = [X_f, X_g] = [\pi^\sharp df, \pi^\sharp dg],$$

and the two minus signs cancel. So the cotangent algebroid of a symplectic manifold is canonically its tangent bundle. This suggests that a symplectic groupoid of (M, ω) is the pair groupoid $M \times M$, which Chapter 3 confirms.

Example 2.4 (The dual of a Lie algebra). Let $\mathfrak{g}^*$ carry the Lie–Poisson structure of Definition 2.4. It is a vector space, so $T_\mu^* \mathfrak{g}^* \cong \mathfrak{g}$ at every point and the cotangent bundle is trivial,

$$T^* \mathfrak{g}^* = \mathfrak{g}^* \times \mathfrak{g}.$$

A one-form is therefore a $\mathfrak{g}$ -valued function on $\mathfrak{g}^*$. The constant ones are the elements of $\mathfrak{g}$ (Section 2.3.1). Each of them is exact, because $d_\mu \ell_\xi = \xi$ for every μ . So the constant section ξ is the form $d\ell_\xi$. These sections span the fibre at every point.

The anchor sends $d\ell_\xi$ to the Hamiltonian field X_{ℓ_ξ} . Its value follows from the Lie–Poisson equation (2.3.1) with $H = \ell_\xi$,

$$\pi^\sharp d\ell_\xi|_\mu = X_{\ell_\xi}(\mu) = -\text{ad}_\xi^* \mu.$$

By Section 2.3.3 this is $a(\xi)_\mu$, the value at μ of the vector field generating the coadjoint action.

The bracket of two constant sections follows from (2.4.1) and $\{\ell_\xi, \ell_\eta\} = \ell_{[\xi, \eta]}$,

$$[d\ell_\xi, d\ell_\eta]_\pi = -d\{\ell_\xi, \ell_\eta\} = -d\ell_{[\xi, \eta]}.$$

So on constant sections the bracket is the bracket of $\mathfrak{g}$ with a minus sign.

Both the anchor and the bracket differ from the action algebroid $\mathfrak{g} \ltimes \mathfrak{g}^*$ of the coadjoint action (Example 1.6), which has anchor $-a$ and bracket $[\xi, \eta]$ on constant sections. One map removes both differences. Put

$$\Phi: T^* \mathfrak{g}^* \rightarrow \mathfrak{g} \ltimes \mathfrak{g}^*, \quad \Phi(\mu, \xi) = (\mu, -\xi).$$

The anchors agree,

$$\rho(\Phi(\mu, \xi)) = -a(-\xi)_\mu = a(\xi)_\mu = \pi^\sharp d\ell_\xi|_\mu,$$

and on constant sections so do the brackets,

$$[\Phi\xi, \Phi\eta] = [-\xi, -\eta] = [\xi, \eta] = \Phi(-[\xi, \eta]) = \Phi([d\ell_\xi, d\ell_\eta]_\pi).$$

Agreement extends from the constant sections to all sections. Write a section in a basis of $\mathfrak{g}$, with coefficients in $C^\infty(\mathfrak{g}^*)$. Both brackets are antisymmetric and expand through the

Leibniz rule of Definition 1.3. In the expansions the coefficients multiply brackets of constant sections, which agree, and the derivative terms contain only the anchors, which agree. So the two expansions are equal, and Φ is an isomorphism of Lie algebroids.

That action algebroid is the algebroid of the action groupoid $G \ltimes \mathfrak{g}^*$ (Example 1.3). This suggests that a groupoid integrating $T^*\mathfrak{g}^*$ is $G \ltimes \mathfrak{g}^*$, which Chapter 3 confirms.

Chapter 3

Symplectic Groupoids

3.1. The definition and its Lagrangian form

Definition 3.1 (Symplectic groupoid). A *symplectic groupoid* is a Lie groupoid $\mathcal{G} \rightrightarrows M$ together with a symplectic form $\Omega \in \Omega^2(\mathcal{G})$ that is *multiplicative*:

$$m^*\Omega = \text{pr}_1^*\Omega + \text{pr}_2^*\Omega \quad \text{on } \mathcal{G}^{(2)}, \quad (3.1.1)$$

where $m: \mathcal{G}^{(2)} \rightarrow \mathcal{G}$ is the multiplication and $\text{pr}_1, \text{pr}_2: \mathcal{G}^{(2)} \rightarrow \mathcal{G}$ are the two projections.

In Section 3.2 we show that $\dim \mathcal{G} = 2 \dim M$, that the units are Lagrangian, and derive other key structure consequences of this relation. On tangent vectors multiplicativity reads

$$\Omega_{gh}(Tm(X, X'), Tm(Y, Y')) = \Omega_g(X, Y) + \Omega_h(X', Y') \quad (3.1.2)$$

for all composable tangent pairs $(X, X'), (Y, Y') \in T_{(g,h)}\mathcal{G}^{(2)}$. A pair $(X, X') \in T_g\mathcal{G} \times T_h\mathcal{G}$ is composable when $Ts(X) = Tt(X')$. Most symplectic forms on a Lie groupoid are not multiplicative.

Proposition 3.1 (Lagrangian form of multiplicativity). *Let $\mathcal{G} \rightrightarrows M$ be a Lie groupoid carrying a symplectic form Ω , and let $\overline{\mathcal{G}}$ denote $\mathcal{G}$ with the symplectic form $-\Omega$. Then Ω is multiplicative if and only if the graph of multiplication*

$$\text{Gr}(m) = \{(g, h, gh) : (g, h) \in \mathcal{G}^{(2)}\}$$

is an isotropic submanifold of $\mathcal{G} \times \mathcal{G} \times \overline{\mathcal{G}}$.

Proof. A tangent vector to $\text{Gr}(m)$ at (g, h, gh) has the form $(X, X', Tm(X, X'))$ for a composable pair (X, X') . The ambient symplectic form is $\text{pr}_1^*\Omega + \text{pr}_2^*\Omega - \text{pr}_3^*\Omega$, so two such tangent vectors pair to

$$\Omega_g(X, Y) + \Omega_h(X', Y') - \Omega_{gh}(Tm(X, X'), Tm(Y, Y')).$$

This vanishes for all composable pairs precisely when (3.1.2) holds. That is, precisely when Ω is multiplicative. $\square$

The proposition states only that $\text{Gr}(m)$ is isotropic. Calling it Lagrangian requires it to have half the dimension of $\mathcal{G} \times \mathcal{G} \times \overline{\mathcal{G}}$. That count needs $\dim \mathcal{G} = 2 \dim M$, which is itself a consequence of multiplicativity. It is proved in Section 3.2 and the statement is completed there in Remark 3.1.

The following two examples run through the chapter: the pair groupoid of a symplectic manifold and the cotangent groupoid $T^*G \rightrightarrows \mathfrak{g}^*$. Their mechanics is the subject of Chapter 4.

Example 3.1 (The pair groupoid of a symplectic manifold). Let (M, ω) be a symplectic manifold and take the pair groupoid $M \times M \rightrightarrows M$ of Example 1.2, with $s(y, x) = x$, $t(y, x) = y$ and $(z, y)(y, x) = (z, x)$. Endow it with the two-form

$$\Omega = \text{pr}_1^* \omega - \text{pr}_2^* \omega, \quad \text{pr}_1(y, x) = y, \quad \text{pr}_2(y, x) = x.$$

It is closed because ω is. For non-degeneracy split the tangent space at (y, x) ,

$$T_{(y,x)}(M \times M) = T_y M \oplus T_x M,$$

and write ω_y and ω_x for the values of ω at the two points. These are non-degenerate bilinear forms on $T_y M$ and on $T_x M$. On tangent vectors

$$\Omega_{(y,x)}((a, b), (a', b')) = \omega_y(a, a') - \omega_x(b, b'). \quad (3.1.3)$$

No term mixes a with b' , so the two summands are Ω -orthogonal and Ω is the sum of one form on each of them. Now let (a, b) be in the kernel. Taking $b' = 0$ gives $\omega_y(a, a') = 0$ for every a' , hence $a = 0$. Taking $a' = 0$ gives $b = 0$ the same way. So Ω is a symplectic form on the manifold $M \times M$. In the notation of Proposition 3.1 this manifold is $M \times \overline{M}$.

It is multiplicative. A composable pair of tangent vectors at $((z, y), (y, x))$ has the form $((c, b), (b, a))$, with $Tm((c, b), (b, a)) = (c, a)$. By (3.1.3)

$$\Omega_{(z,y)}((c, b), (c', b')) + \Omega_{(y,x)}((b, a), (b', a')) = \omega_z(c, c') - \omega_x(a, a') = \Omega_{(z,x)}((c, a), (c', a')),$$

the middle point cancelling. This is (3.1.2). Hence $(M \times M, \Omega) \rightrightarrows M$ is a symplectic groupoid.

The structural features derived in general in the next section can be checked directly here. The units form the diagonal $\Delta = \{(x, x)\}$, on which Ω vanishes,

$$\Omega_{(x,x)}((a, a), (a', a')) = \omega_x(a, a') - \omega_x(a, a') = 0.$$

So Δ is Lagrangian, of dimension $\dim M = \frac{1}{2} \dim(M \times M)$. The inversion $(y, x) \mapsto (x, y)$ swaps the two factors and so pulls Ω back to $-\Omega$. The source and target fibres through (x, x) , namely $M \times \{x\}$ and $\{x\} \times M$, are Ω -orthogonal.

Example 3.2 (The dual of a Lie algebra). Let G be a Lie group with Lie algebra $\mathfrak{g}$. Its dual $\mathfrak{g}^*$ carries the Lie–Poisson structure of Section 2.3. This is an example of a base that is not symplectic. We ask what integrates it. Chapter 2 has already fixed the answer and we only read it off.

The groupoid. By Example 2.4 the cotangent algebroid of $\mathfrak{g}^*$ is the action algebroid of the coadjoint action, and by Example 1.6 that is the algebroid of the action groupoid $G \ltimes \mathfrak{g}^*$. So the groupoid is already chosen. Substituting the coadjoint action into Example 1.3 gives its structure maps,

$$s(g, \mu) = \mu, \quad t(g, \mu) = \text{Ad}_g^* \mu, \quad u(\mu) = (e, \mu),$$

with $\iota(g, \mu) = (g^{-1}, \text{Ad}_g^* \mu)$ and $(h, \text{Ad}_g^* \mu)(g, \mu) = (hg, \mu)$. Nothing here is new, so nothing needs checking.

The manifold. So far the arrows are only the product $G \times \mathfrak{g}^*$. That product is a trivialization of a cotangent bundle. Take $\alpha_g \in T_g^* G$ and record it at the identity by left translation,

$$\langle \mu, \xi \rangle = \langle \alpha_g, T_e L_g \xi \rangle, \quad \xi \in \mathfrak{g}. \quad (3.1.4)$$

The map $\alpha_g \mapsto (g, \mu)$ is a diffeomorphism $T^* G \cong G \times \mathfrak{g}^*$ called the *left trivialisation*. From now on we write arrows as pairs (g, μ) and use the canonical symplectic form of $T^* G$.

Recording α_g by right translation instead gives ν with $\langle \nu, \xi \rangle = \langle \alpha_g, T_e R_g \xi \rangle$. The two records are related. Conjugation $C_{g^{-1}}(h) = g^{-1} h g$ satisfies $L_g \circ C_{g^{-1}} = R_g$, so $T_e R_g = T_e L_g \circ \text{Ad}_{g^{-1}}$ and therefore

$$\langle \nu, \xi \rangle = \langle \mu, \text{Ad}_{g^{-1}} \xi \rangle = \langle \text{Ad}_g^* \mu, \xi \rangle.$$

So $\nu = \text{Ad}_g^* \mu$. Compare this with the structure maps above. An arrow is one covector on G . Its source and its target are the two ways of carrying that covector to the identity.

The canonical form. Let θ^L be the left Maurer–Cartan form, the $\mathfrak{g}$ -valued one-form

$$\theta_g^L(v) = T_g L_{g^{-1}} v \in \mathfrak{g}, \quad v \in T_g G,$$

which undoes left translation. In the trivialisation the tautological one-form θ of T^*G becomes a pairing. Let $q: T^*G \rightarrow G$ be the bundle projection. We do not call it π , which is reserved for Poisson structures. Let w be a tangent vector to T^*G at the point α_g , written (g, μ) in the trivialisation, and put $v = Tq(w) \in T_g G$. The tautological form is defined by $\theta(w) = \langle \alpha_g, v \rangle$. Undoing and then reapplying the left translation of v gives $v = T_e L_g(\theta_g^L v)$, so

$$\theta(w) = \langle \alpha_g, v \rangle = \langle \alpha_g, T_e L_g(\theta_g^L v) \rangle = \langle \mu, \theta_g^L v \rangle,$$

the last step by (3.1.4) with $\xi = \theta_g^L v$. The vector w was arbitrary, so at every point (g, μ) of T^*G

$$\theta_{(g, \mu)} = \langle \mu, \theta_g^L \rangle. \quad (3.1.5)$$

The symplectic form is $\Omega = -d\theta$.

Multiplicativity. We show that θ satisfies (3.1.1). Applying $-d$ then gives the same for Ω , because d commutes with pullback.

First we coordinatise the composable pairs. A pair $((h, \nu), (g, \mu))$ is composable when $s(h, \nu) = t(g, \mu)$, that is when $\nu = \text{Ad}_g^* \mu$. So ν is not free and the pair is determined by h, g and μ . The map

$$G \times G \times \mathfrak{g}^* \longrightarrow (T^*G)^{(2)}, \quad (h, g, \mu) \longmapsto ((h, \text{Ad}_g^* \mu), (g, \mu)),$$

is a diffeomorphism. In these coordinates

$$\text{pr}_1(h, g, \mu) = (h, \text{Ad}_g^* \mu), \quad \text{pr}_2(h, g, \mu) = (g, \mu), \quad m(h, g, \mu) = (hg, \mu).$$

Next we need the differential of the multiplication of G . Write $\text{mult}: G \times G \rightarrow G$ for it. Then

$$T_{(h, g)} \text{mult}(v, w) = T_h R_g v + T_g L_h w, \quad v \in T_h G, \quad w \in T_g G. \quad (3.1.6)$$

The differential is linear on $T_h G \oplus T_g G$, so it is enough to take the two summands separately. Freezing the second argument gives $\text{mult}(\cdot, g) = R_g$, whose differential at h is $T_h R_g$. Freezing the first gives $\text{mult}(h, \cdot) = L_h$, whose differential at g is $T_g L_h$.

The Maurer–Cartan form satisfies

$$\theta_{hg}^L = \text{Ad}_{g^{-1}} \theta_h^L + \theta_g^L. \quad (3.1.7)$$

To prove it, put (3.1.6) into the definition of θ^L at hg ,

$$\theta_{hg}^L(T \text{mult}(v, w)) = T_{hg} L_{(hg)^{-1}} T_h R_g v + T_{hg} L_{(hg)^{-1}} T_g L_h w.$$

The second term is $T_g L_{g^{-1}} w = \theta_g^L(w)$. Take the first term. Left and right translations commute, so $L_{g^{-1}h^{-1}} \circ R_g = L_{g^{-1}} \circ R_g \circ L_{h^{-1}} = C_{g^{-1}} \circ L_{h^{-1}}$. Differentiate this composite at h . It sends v first to $T_h L_{h^{-1}} v = \theta_h^L(v)$, which lies in $T_e G = \mathfrak{g}$, and then through $T_e C_{g^{-1}} = \text{Ad}_{g^{-1}}$. So the first term is $\text{Ad}_{g^{-1}}(\theta_h^L(v))$, which proves (3.1.7).

Now fix a composable pair (h, g, μ) and a tangent vector (v, w, λ) at it, with $v \in T_h G$, $w \in T_g G$ and $\lambda \in \mathfrak{g}^*$. Multiplication sends it to $Tm(v, w, \lambda) = (T \text{mult}(v, w), \lambda)$. Evaluate (3.1.5) at the point (hg, μ) and then use (3.1.7) and the definition of Ad^* ,

$$(m^* \theta)_{(h, g, \mu)}(v, w, \lambda) = \langle \mu, \theta_{hg}^L(T \text{mult}(v, w)) \rangle = \langle \text{Ad}_g^* \mu, \theta_h^L(v) \rangle + \langle \mu, \theta_g^L(w) \rangle.$$

Read the two terms through (3.1.5) again. The first is θ at the point $(h, \text{Ad}_g^* \mu)$ evaluated on v . That point is the first arrow, so the first term is $(\text{pr}_1^* \theta)(v, w, \lambda)$. The second is θ at (g, μ) evaluated on w , so it is $(\text{pr}_2^* \theta)(v, w, \lambda)$. The tangent vector was arbitrary, so

$$m^* \theta = \text{pr}_1^* \theta + \text{pr}_2^* \theta$$

and $(T^*G, \Omega) \rightrightarrows \mathfrak{g}^*$ is a symplectic groupoid. The construction is due to Coste, Dazord and Weinstein [10].

What the arrows mean. Take $G = \mathrm{SO}(3)$ and the free rigid body of Section 2.3.4. Its configuration is a rotation, so its configuration space is a group. The phase space is $T^*\mathrm{SO}(3)$ of dimension six, and it is symplectic. The energy does not depend on the orientation. It is therefore invariant under the rotations and the dynamics descends to the quotient. The quotient has dimension three and its coordinate is the body momentum μ . Three is odd, so the quotient cannot be symplectic. What survives on it is a degenerate Poisson bracket. This is where the Poisson geometry of Chapter 2 comes from.

The groupoid gives the three dropped dimensions back. They are the group element of the arrow (g, μ) . The symplectic groupoid over the reduced phase space is the unreduced phase space. The source of (g, μ) is the momentum in the body frame and the target is the same momentum in the space frame, by the two records above and by Section 2.3.4. The rotation g is what carries one frame to the other. So an arrow is a change of frame, and composing arrows composes changes of frame. Chapter 4 performs the reduction by reading s and t .

3.2. Structural consequences of multiplicativity

Multiplicativity is a strong condition. This section presents four consequences of it. The dimension of $\mathcal{G}$ is fixed by the base. The unit submanifold is Lagrangian. The inversion is anti-symplectic. The source and target fibres are symplectic orthogonal complements of each other at every point. The rest of the chapter uses all four. These fundamental facts are due to Coste, Dazord and Weinstein [10]. Modern treatments are in [4, 6].

Three facts about a symplectic vector space are needed. For a linear subspace $W \subseteq T_g\mathcal{G}$ we write

$$W^\Omega = \{ v \in T_g\mathcal{G} : \Omega_g(v, w) = 0 \text{ for all } w \in W \}$$

for its *symplectic orthogonal complement*. The first two facts are $\dim W + \dim W^\Omega = \dim \mathcal{G}$ and $(W^\Omega)^\Omega = W$. The third follows from the first. An isotropic subspace satisfies $W \subseteq W^\Omega$. So $\dim W \leq \dim \mathcal{G} - \dim W$ and the dimension of W is at most $\frac{1}{2} \dim \mathcal{G}$.

Throughout the section $(\mathcal{G} \rightrightarrows M, \Omega)$ is a symplectic groupoid.

We first record how the tangent multiplication acts on a pair with one zero entry. Nothing here uses Ω , and the statement holds in any Lie groupoid. Let $(g, h) \in \mathcal{G}^{(2)}$ and let $X \in \ker T_g s$. Then $Ts(X) = 0$, so the pair $(X, 0)$ is composable. The kernel $\ker T_g s$ is the tangent space at g to the source fibre $\mathcal{G}_{s(g)}$, so there is a curve γ inside that fibre with $\gamma(0) = g$ and $\gamma'(0) = X$. Along it $s(\gamma(\tau)) = s(g) = t(h)$, so every pair $(\gamma(\tau), h)$ is composable and $m(\gamma(\tau), h) = R_h(\gamma(\tau))$. A curve inside the target fibre $\mathcal{G}^{t(h)}$ through h gives the mirror statement for the left translation. Differentiating both at $\tau = 0$ gives

$$Tm(X, 0) = T_g R_h(X), \quad Tm(0, Y) = T_h L_g(Y), \quad (3.2.1)$$

for $X \in \ker T_g s$ and $Y \in \ker T_h t$, where 0 is the zero tangent vector at the arrow in question.

Lemma 3.1 (Three products). *Let $(\mathcal{G} \rightrightarrows M, \Omega)$ be a symplectic groupoid.*

- (i) *The unit submanifold $u(M)$ is isotropic. For $x \in M$ and $a, b \in T_x M$ one has $\Omega_{1_x}(Tu a, Tu b) = 0$.*
- (ii) *The source fibre and the target fibre through an arrow are Ω -orthogonal. For $g \in \mathcal{G}$, $X \in \ker T_g s$ and $Y \in \ker T_g t$ one has $\Omega_g(X, Y) = 0$, that is $\ker T_g t \subseteq (\ker T_g s)^\Omega$.*

(iii) *The inversion is an anti-symplectomorphism, $\iota^*\Omega = -\Omega$. It carries Lagrangian submanifolds of $(\mathcal{G}, \Omega)$ to Lagrangian submanifolds.*

Proof. Each part evaluates (3.1.2) on a composable pair whose product is already known. The three products are $1_x 1_x = 1_x$, $g 1_x = g$ and $g g^{-1} = 1_{t(g)}$.

(i). From $s \circ u = t \circ u = \text{id}_M$ we get $Ts(Tu a) = Tt(Tu a) = a$, so the pair $(Tu a, Tu a)$ is composable. The map $x \mapsto (1_x, 1_x)$ sends M into $\mathcal{G}^{(2)}$ and satisfies $m \circ (u, u) = u$. Differentiating that identity gives $Tm(Tu a, Tu a) = Tu(a)$, and the same for b . Hence

$$\begin{aligned}\Omega_{1_x}(Tu a, Tu b) &= \Omega_{1_x}(Tm(Tu a, Tu a), Tm(Tu b, Tu b)) \\ &= \Omega_{1_x}(Tu a, Tu b) + \Omega_{1_x}(Tu a, Tu b).\end{aligned}$$

A number equal to twice itself is zero.

(ii). Put $x = s(g)$ and use the pair $(g, 1_x)$, whose product is g . The right translation R_{1_x} is the identity of the source fibre $\mathcal{G}_x$, so the first identity of (3.2.1) gives $Tm(X, 0) = X$. The left translation $L_g: \mathcal{G}^x \rightarrow \mathcal{G}^{t(g)}$ is a diffeomorphism of target fibres with $L_g(1_x) = g$, and its differential carries $\ker T_{1_x}t$ isomorphically onto $\ker T_g t$. So $Y = T_{1_x}L_g(\eta)$ for a unique $\eta \in \ker T_{1_x}t$, and the second identity gives $Tm(0, \eta) = Y$. Then

$$\Omega_g(X, Y) = \Omega_g(Tm(X, 0), Tm(0, \eta)) = \Omega_g(X, 0) + \Omega_{1_x}(0, \eta) = 0.$$

(iii). Fix $g \in \mathcal{G}$ and $X, Y \in T_g \mathcal{G}$. From $t \circ \iota = s$ we get $Tt(T\iota X) = Ts(X)$, so the pair $(X, T\iota X)$ is composable, and the same holds for $(Y, T\iota Y)$. Differentiating $m \circ (\text{id}, \iota) = u \circ t$ gives

$$Tm(X, T\iota X) = Tu(Tt X) \in Tu(T_{t(g)}M),$$

and likewise for Y . So

$$\Omega_{1_{t(g)}}(Tu Tt X, Tu Tt Y) = \Omega_g(X, Y) + \Omega_{g^{-1}}(T\iota X, T\iota Y).$$

The vectors $Tt X$ and $Tt Y$ lie in $T_{t(g)}M$, so the left side vanishes by part (i). This gives $\iota^*\Omega = -\Omega$. A submanifold is Lagrangian for Ω exactly when it is Lagrangian for $-\Omega$, which gives the last claim. $\square$

Proposition 3.2 (Structure of a symplectic groupoid). *Let $(\mathcal{G} \rightrightarrows M, \Omega)$ be a symplectic groupoid.*

(i) $\dim \mathcal{G} = 2 \dim M$.

(ii) *The unit submanifold $u(M)$ is Lagrangian.*

(iii) *For every $g \in \mathcal{G}$,*

$$(\ker T_g s)^\Omega = \ker T_g t, \quad (\ker T_g t)^\Omega = \ker T_g s.$$

Proof. Write $n = \dim M$ and $N = \dim \mathcal{G}$.

Take $g = 1_x$ in Lemma 3.1(ii), which gives $\ker T_{1_x}t \subseteq (\ker T_{1_x}s)^\Omega$. Both s and t are submersions, so both kernels have dimension $N - n$. The complement has dimension n , hence $N - n \leq n$, that is $N \leq 2n$.

By Lemma 3.1(i) the subspace $Tu(T_x M)$ is isotropic. Its dimension is n , because $Ts \circ Tu = \text{id}$ makes Tu injective. An isotropic subspace has dimension at most $\frac{1}{2}N$. Hence $n \leq \frac{1}{2}N$, that is $N \geq 2n$. The two bounds meet and $N = 2n$, which gives (i).

Part (ii) follows directly. The subspace $Tu(T_x M)$ is isotropic of dimension $\frac{1}{2}N$ and so is Lagrangian.

For part (iii) fix g . By (i) the kernel $\ker T_g t$ has dimension $N - n = n$, and $(\ker T_g s)^\Omega$ has dimension $N - (N - n) = n$ as well. Equal dimensions turn the inclusion of Lemma 3.1(ii) into an equality. Taking Ω -orthogonal complements of both sides and using $(W^\Omega)^\Omega = W$ gives the second identity. $\square$

Remark 3.1. Proposition 3.1 showed that the graph of the multiplication is isotropic. Now it is Lagrangian. The composable pairs form the fibre product of two submersions. So $\mathcal{G}^{(2)}$ has codimension $\dim M$ in $\mathcal{G} \times \mathcal{G}$ and

$$\dim \mathcal{G}^{(2)} = 2 \dim \mathcal{G} - \dim M = \frac{3}{2} \dim \mathcal{G},$$

which is half of $\dim(\mathcal{G} \times \mathcal{G} \times \overline{\mathcal{G}}) = 3 \dim \mathcal{G}$. The multiplication of a symplectic groupoid is given by a Lagrangian submanifold of $\mathcal{G} \times \mathcal{G} \times \overline{\mathcal{G}}$.

3.3. The Poisson structure of the base

A symplectic groupoid determines a Poisson structure on its base. The theorem is due to Weinstein [9]. The detailed account is of Coste, Dazord and Weinstein [10]. Modern treatments are in [4, 6].

We write $\{\cdot, \cdot\}_\Omega$ for the Poisson bracket of $(\mathcal{G}, \Omega)$ (Example 2.1). Its Hamiltonian fields are fixed by $\iota_{X_\phi} \Omega = d\phi$. So $X_\phi \psi = \{\psi, \phi\}$ and $\{\phi, \psi\}_\Omega = \Omega(X_\phi, X_\psi)$.

The whole section studies two assignments. A function on M has two pullbacks to $\mathcal{G}$, and each pullback has a Hamiltonian vector field,

$$f \longmapsto X_{t^*f}, \quad f \longmapsto X_{s^*f}, \quad f \in C^\infty(M).$$

Lemma 3.2 collects three properties of these fields, and every later result of the section applies one of them. The inversion satisfies $\iota^* \Omega = -\Omega$ by Lemma 3.1(iii). It exchanges s with t and reverses products, so it exchanges left translation with right translation. Every statement about one assignment therefore has a mirror about the other. We state both sides and prove one of each pair.

Three pieces of bookkeeping about the tangent multiplication are needed. Nothing in them uses Ω . First, on $\mathcal{G}^{(2)}$ one has $t \circ m = t \circ \text{pr}_1$ and $s \circ m = s \circ \text{pr}_2$, because $t(gh) = t(g)$ and $s(gh) = s(h)$. Differentiating gives

$$Tt(Tm(v, v')) = Tt(v), \quad Ts(Tm(v, v')) = Ts(v') \quad (3.3.1)$$

for every composable pair $(v, v') \in T_{(g,h)}\mathcal{G}^{(2)}$. Second, left translation by g is a diffeomorphism between target fibres, so $T_h L_g$ carries $\ker T_h t$ onto $\ker T_{gh} t$. By (3.2.1) it does so as $y \mapsto Tm(0, y)$. Third, the differential $T_{(g,h)} m: T_{(g,h)}\mathcal{G}^{(2)} \rightarrow T_{gh}\mathcal{G}$ is surjective. To see this take $w \in T_{gh}\mathcal{G}$. The target map is a submersion, so there is $v \in T_g\mathcal{G}$ with $Tt(v) = Tt(w)$, and there is $v' \in T_h\mathcal{G}$ with $Tt(v') = Ts(v)$. The pair (v, v') is composable, and $Tt(Tm(v, v')) = Tt(v) = Tt(w)$ by (3.3.1). So $w - Tm(v, v')$ lies in $\ker T_{gh} t$, which is contained in the image of $T_{(g,h)} m$ by the second point. Hence w lies in the image as well.

Lemma 3.2 (Pullbacks and their Hamiltonian fields). *Let $(\mathcal{G} \rightrightarrows M, \Omega)$ be a symplectic groupoid and let $f_1, f_2 \in C^\infty(M)$.*

(i) *For every $g \in \mathcal{G}$,*

$$\ker T_g s = \{ X_{t^*f_1}(g) : f_1 \in C^\infty(M) \}, \quad \ker T_g t = \{ X_{s^*f_1}(g) : f_1 \in C^\infty(M) \}.$$

(ii) *The field $X_{t^*f_1}$ is right-invariant and the field $X_{s^*f_1}$ is left-invariant. For every $(g, h) \in \mathcal{G}^{(2)}$,*

$$T_g R_h(X_{t^*f_1}(g)) = X_{t^*f_1}(gh), \quad T_h L_g(X_{s^*f_1}(h)) = X_{s^*f_1}(gh).$$

(iii) $\{s^*f_1, t^*f_2\}_\Omega = 0$.

Proof. (i). Fix g and put $y = t(g)$. The differential of a target pullback is $d(t^*f_1)_g = d(f_1)_y \circ T_g t$. The target map is a submersion, so $T_g t$ is onto and $\alpha \mapsto \alpha \circ T_g t$ is an injection of $T_y^* M$ into

$T_g^*\mathcal{G}$ with image the annihilator of $\ker T_g t$. Every covector at y is $d(f_1)_y$ for some f_1 , so the covectors $d(t^*f_1)_g$ fill that annihilator. The form Ω_g is non-degenerate. The map $v \mapsto \Omega_g(v, \cdot)$ carries $(\ker T_g t)^\Omega$ onto the annihilator of $\ker T_g t$. Since $\Omega_g(X_{t^*f_1}(g), \cdot) = d(t^*f_1)_g$, the vectors $X_{t^*f_1}(g)$ fill $(\ker T_g t)^\Omega$, which is $\ker T_g s$ by Proposition 3.2(iii). Exchanging s and t gives the second identity.

(ii). Put $X = X_{t^*f_1}(g)$. By (i) we have $X \in \ker T_g s$. So the pair $(X, 0)$ at (g, h) is composable and $Tm(X, 0) = T_g R_h(X)$ by (3.2.1). Let $w \in T_{gh}\mathcal{G}$ and write $w = Tm(v, v')$ with (v, v') composable, which the bookkeeping above allows. Multiplicativity (3.1.2) gives

$$\Omega_{gh}(Tm(X, 0), Tm(v, v')) = \Omega_g(X, v) + \Omega_h(0, v') = d(t^*f_1)_g(v).$$

By (3.3.1) we have $Tt(w) = Tt(v)$, so

$$d(t^*f_1)_{gh}(w) = df_1(Tt w) = df_1(Tt v) = d(t^*f_1)_g(v).$$

Both quantities on the left depend on w alone, so the choice of the pair (v, v') plays no role. The two right-hand sides agree, hence $\Omega_{gh}(T_g R_h(X), w) = d(t^*f_1)_{gh}(w)$ for every $w \in T_{gh}\mathcal{G}$. This is the equation that defines $X_{t^*f_1}(gh)$, and the non-degeneracy of Ω_{gh} admits only one solution. So $T_g R_h(X) = X_{t^*f_1}(gh)$. The identity for $X_{s^*f_1}$ is the same computation with s and t exchanged, using $Tm(0, y) = T_h L_g(y)$ in place of $Tm(X, 0) = T_g R_h(X)$.

(iii). By (i) the field $X_{t^*f_2}$ takes values in $\ker Ts$, and the differential $d(s^*f_1) = df_1 \circ Ts$ annihilates $\ker Ts$. So

$$\{s^*f_1, t^*f_2\}_\Omega = X_{t^*f_2}(s^*f_1) = d(s^*f_1)(X_{t^*f_2}) = 0.$$

□

Theorem 3.1 (Weinstein, Coste–Dazord–Weinstein). *Let $(\mathcal{G} \rightrightarrows M, \Omega)$ be a symplectic groupoid. There is exactly one Poisson structure π on M for which the target map $t: (\mathcal{G}, \Omega) \rightarrow (M, \pi)$ is a Poisson map. Its bracket is determined by*

$$t^*\{f_1, f_2\} = \{t^*f_1, t^*f_2\}_\Omega, \quad f_1, f_2 \in C^\infty(M). \quad (3.3.2)$$

The source map is a Poisson map from $(\mathcal{G}, \Omega)$ to $\overline{M} = (M, -\pi)$.

Proof. The bracket of two target pullbacks is constant on each target fibre. Fix $f_1, f_2 \in C^\infty(M)$ and let $(g, h) \in \mathcal{G}^{(2)}$. Put $X_i = X_{t^*f_i}(g)$, which lies in $\ker T_g s$ by Lemma 3.2(i). By part (ii) of the same lemma and by (3.2.1),

$$X_{t^*f_i}(gh) = T_g R_h(X_i) = Tm(X_i, 0),$$

so multiplicativity (3.1.2) gives

$$\{t^*f_1, t^*f_2\}_\Omega(gh) = \Omega_{gh}(Tm(X_1, 0), Tm(X_2, 0)) = \Omega_g(X_1, X_2) = \{t^*f_1, t^*f_2\}_\Omega(g).$$

Now take $g = 1_{t(h)}$. The pair $(1_{t(h)}, h)$ is composable and $1_{t(h)}h = h$, so for every $h \in \mathcal{G}$

$$\{t^*f_1, t^*f_2\}_\Omega(h) = \{t^*f_1, t^*f_2\}_\Omega(1_{t(h)}).$$

Put $\{f_1, f_2\} = u^*\{t^*f_1, t^*f_2\}_\Omega$, a smooth function on M . Its pullback along t takes at h the value $\{t^*f_1, t^*f_2\}_\Omega(1_{t(h)})$. This is the identity (3.3.2).

The bracket is Poisson. The target map is surjective, so t^* is injective. Bilinearity is clear from (3.3.2). For the Jacobi identity apply (3.3.2) twice,

$$t^*\{\{f_1, f_2\}, f_3\} = \{t^*\{f_1, f_2\}, t^*f_3\}_\Omega = \{\{t^*f_1, t^*f_2\}_\Omega, t^*f_3\}_\Omega,$$

and add the two identities obtained by cycling f_1, f_2 and f_3 . The right side becomes the Jacobi expression of $\{\cdot, \cdot\}_\Omega$ on t^*f_1, t^*f_2 and t^*f_3 , which is zero. So t^* of the Jacobi expression on M is zero, and it vanishes because t^* is injective. Antisymmetry and the Leibniz rule follow in the same way from one application of (3.3.2) instead of two.

Uniqueness. A Poisson structure for which t is a Poisson map satisfies (3.3.2), and that determines every bracket $\{f_1, f_2\}$ because t^* is injective.

The source map. Write $s = t \circ \iota$ and use $\iota^*\Omega = -\Omega$ from Lemma 3.1(iii). A symplectic form and its negative have brackets differing by a sign, so $\{\iota^*A, \iota^*B\}_\Omega = -\iota^*\{A, B\}_\Omega$ for $A, B \in C^\infty(\mathcal{G})$. Hence

$$s^*\{f_1, f_2\} = \iota^*t^*\{f_1, f_2\} = \iota^*\{t^*f_1, t^*f_2\}_\Omega = -\{s^*f_1, s^*f_2\}_\Omega,$$

which is (3.3.2) for s and $-\pi$. $\square$

We call π the induced structure. Everything later in this chapter and in Chapter 4 rests on the identity (3.3.2).

Remark 3.2 (Conventions). Sources differ over which structure map is the Poisson one. Weinstein states the theorem with the source map Poisson and the target map anti-Poisson [9], and the lectures of Crainic and Fernandes do the same [15]. We use the opposite orientation, which is the one used in the work on generating functions and Poisson integrators followed in Chapter 4 [26]. A reader comparing statements with the literature should check the orientation first.

Definition 3.2 (Symplectic realisation). Let (M, π) be a Poisson manifold. A *symplectic realisation* of (M, π) is a symplectic manifold together with a surjective submersion onto M which is a Poisson map.

By Theorem 3.1 the target map of a symplectic groupoid is a symplectic realisation of (M, π) and the source map is one of $\overline{M}$,

$$M \xleftarrow{t} \mathcal{G} \xrightarrow{s} \overline{M}.$$

The fibres of the two maps are symplectic orthogonal complements of each other by Proposition 3.2(iii). A pair of realisations of opposite sign with this orthogonality is called a *dual pair*. Weinstein obtained Theorem 3.1 from the theory of dual pairs [9, 4]. The two structure maps and the orthogonality of their fibres are the starting point of the integrator constructions in Chapter 4.

Corollary 3.1 (Lifted dynamics). *Let $(\mathcal{G} \rightrightarrows M, \Omega)$ be a symplectic groupoid, let π be the induced structure, let $H \in C^\infty(M)$ and let φ_τ be the flow of X_H on (M, π) .*

(i) *The field X_{t^*H} is tangent to the source fibres and*

$$Tt(X_{t^*H}) = X_H \circ t.$$

Its flow $\widetilde{\varphi}_\tau$ satisfies $s \circ \widetilde{\varphi}_\tau = s$ and $t \circ \widetilde{\varphi}_\tau = \varphi_\tau \circ t$ wherever it is defined.

(ii) *The field X_{s^*H} is tangent to the target fibres and*

$$Ts(X_{s^*H}) = -X_H \circ s.$$

Its flow $\widehat{\varphi}_\tau$ satisfies $t \circ \widehat{\varphi}_\tau = t$ and $s \circ \widehat{\varphi}_\tau = \varphi_{-\tau} \circ s$ wherever it is defined.

Proof. The two tangency statements are Lemma 3.2(i). For the projection in (i) test against an arbitrary $f_1 \in C^\infty(M)$. Use $d(t^*f_1) = df_1 \circ Tt$,

$$\langle df_1, Tt X_{t^*H} \rangle = X_{t^*H}(t^*f_1) = \{t^*f_1, t^*H\}_\Omega = t^*\{f_1, H\} = \langle df_1, X_H \rangle \circ t,$$

by (3.3.2) and $\{f_1, H\} = X_H f_1$ (Section 2.2.1). The covector df_1 is arbitrary, so the two tangent vectors agree. In (ii) the same computation runs with s in place of t . The middle step $\{s^*f_1, s^*H\}_\Omega = -s^*\{f_1, H\}$ carries a minus sign, because s is a Poisson map onto $\overline{M}$.

For the flows, differentiate along them. In (i) the curve $\tau \mapsto s(\tilde{\varphi}_\tau(g))$ has velocity $Ts(X_{t^*H}) = 0$, so it stays at $s(g)$. The curve $\tau \mapsto t(\tilde{\varphi}_\tau(g))$ has velocity X_H at its current point. So it is the integral curve of X_H through $t(g)$, and integral curves through a point are unique. In (ii) the same argument applies with $-X_H$ in place of X_H . $\square$

Start the flow of X_{s^*H} at the units. By (ii) the targets stay fixed and the sources move along the flow of H . So the exact solution of the Hamiltonian system appears on $\mathcal{G}$ as a curve of arrows through the units. Section 3.4 turns this curve into so-called bisections, and Chapter 4 obtains numerical methods by replacing it with a computable curve [18].

Proposition 3.3 (Leaves, orbits and Casimirs). *Let $(\mathcal{G} \rightrightarrows M, \Omega)$ be a symplectic groupoid and let π be the induced structure.*

(i) *For every $g \in \mathcal{G}$,*

$$\text{im } \pi_{t(g)}^\# = T_g t(\ker T_g s).$$

(ii) *If the source fibres of $\mathcal{G}$ are connected, the symplectic leaves of (M, π) are the orbits of $\mathcal{G}$.*

(iii) *If the source fibres of $\mathcal{G}$ are connected, a function $C \in C^\infty(M)$ is a Casimir of π if and only if $s^*C = t^*C$.*

Proof. (i). By Lemma 3.2(i) the space $\ker T_g s$ consists of the values $X_{t^*f_1}(g)$ with $f_1 \in C^\infty(M)$. By Corollary 3.1(i),

$$T_g t(X_{t^*f_1}(g)) = X_{f_1}(t(g)) = \pi^\#(d(f_1)_{t(g)}).$$

Every covector at $t(g)$ is $d(f_1)_{t(g)}$ for some f_1 . So as f_1 varies the right side fills $\text{im } \pi_{t(g)}^\#$.

(ii). Recall $\mathcal{O}_x = t(\mathcal{G}_x)$ from Section 1.2.2. Take $x \in M$ and $g \in \mathcal{G}_x$. The source fibre $\mathcal{G}_x$ is connected, so there is a smooth path γ inside $\mathcal{G}_x$ from 1_x to g . The path $t \circ \gamma$ runs from x to $t(g)$ in M . Its velocity is $Tt(\gamma')$ with $\gamma' \in \ker Ts$, so by (i) the velocity lies in the characteristic distribution at every time. A path with this property stays inside one symplectic leaf [7, 6]. So $\mathcal{O}_x$ is contained in the leaf through x .

For the reverse inclusion take $H \in C^\infty(M)$ with flow φ_τ on M . By Corollary 3.1(ii) the arrow $\widehat{\varphi}_\tau(1_x)$ has target $t(1_x) = x$ and source $\varphi_{-\tau}(x)$, so $\varphi_{-\tau}(x)$ lies in $\mathcal{O}_x$ for small τ . Apply this at an arbitrary point of the curve. For every τ_0 in the interval of definition, all points $\varphi_{-\tau}(x)$ with τ near τ_0 lie in the orbit of $\varphi_{-\tau_0}(x)$. Orbits are disjoint, so the set of times τ with $\varphi_{-\tau}(x) \in \mathcal{O}_x$ is open and closed in that interval. It contains $\tau = 0$, so the whole curve stays in $\mathcal{O}_x$. Two points of one leaf are joined by a chain of Hamiltonian flows by Theorem 2.1(i), and each flow keeps the point in its orbit. So the leaf through x is contained in $\mathcal{O}_x$.

(iii). Let C be a Casimir. Then $\{C, f_1\} = 0$ for every f_1 , so C is constant along every Hamiltonian flow, and by Theorem 2.1(i) it is constant on each leaf. By (ii) it is constant on each orbit. For $g \in \mathcal{G}$ the points $s(g)$ and $t(g)$ lie in one orbit, so $C(s(g)) = C(t(g))$.

Conversely let $s^*C = t^*C$. Then $X_{s^*C} = X_{t^*C}$, and by Lemma 3.2(i) this field lies in $\ker Tt$ and in $\ker Ts$. Its image under Tt is $X_C \circ t$ by Corollary 3.1(i), and that image is zero. The target map is onto, so $X_C = 0$ and C is a Casimir by Section 2.2.2. $\square$

The hypothesis in parts (ii) and (iii) is not restrictive. Every Lie groupoid contains an open subgroupoid over the same base whose source fibres are the connected components of the source fibres through the units, and it has the same Lie algebroid [2]. Restricting Ω makes it a symplectic groupoid again, and the induced structure does not change, because it is computed on the units. The inversion carries source fibres onto target fibres, so the

hypothesis holds for the target fibres as well. In Chapter 4 parts (ii) and (iii) are used to derive the conservation laws of the numerical methods.

Example 3.3 (The base of the pair groupoid). Let (M, ω) be a symplectic manifold and take $(M \times M, \Omega) \rightrightarrows M$ of Example 3.1, with $t = \text{pr}_1$, $s = \text{pr}_2$ and $\Omega = \text{pr}_1^* \omega - \text{pr}_2^* \omega$. Solve $\iota_X \Omega = d(t^* f_1)$ for $X = (X_1, X_2)$. By (3.1.3) this reads

$$\omega(X_1, a) - \omega(X_2, b) = df_1(a) \quad \text{for all } (a, b),$$

so $X_1 = X_{f_1}$ and $X_2 = 0$. Hence $X_{t^* f_1} = (X_{f_1}, 0)$ and

$$\{t^* f_1, t^* f_2\}_\Omega = \Omega((X_{f_1}, 0), (X_{f_2}, 0)) = \omega(X_{f_1}, X_{f_2}) = t^* \{f_1, f_2\}_\omega,$$

with $\{\cdot, \cdot\}_\omega$ the bracket of (M, ω) (Example 2.1). By (3.3.2) the structure induced on the base is $\{\cdot, \cdot\}_\omega$ itself.

Example 3.4 (The base of the cotangent groupoid). Take $T^*G \rightrightarrows \mathfrak{g}^*$ of Example 3.2, with $s(g, \mu) = \mu$, $t(g, \mu) = \text{Ad}_g^* \mu$ and the tautological form (3.1.5). It is enough to compute (3.3.2) on the linear functions $\ell_\xi(\mu) = \langle \mu, \xi \rangle$, because $d_\mu \ell_\xi = \xi$ and these differentials fill the cotangent space at every point.

For $h \in G$ put $\Lambda_h(g, \mu) = (hg, \mu)$. The Maurer–Cartan form satisfies $\theta_{hg}^L \circ T_g L_h = T_{hg} L_{(hg)^{-1}} \circ T_g L_h = T_g L_{g^{-1}} = \theta_g^L$, and Λ_h leaves μ unchanged, so $\Lambda_h^* \theta = \theta$. Let V_ξ generate $\tau \mapsto \Lambda_{\exp(\tau\xi)}$, that is $V_\xi(g, \mu) = (T_e R_g \xi, 0)$. Since $\mathcal{L}_{V_\xi} \theta = 0$ and $\Omega = -d\theta$, Cartan’s formula gives $\iota_{V_\xi} \Omega = d(\theta(V_\xi))$. Using $T_e R_g = T_e L_g \circ \text{Ad}_{g^{-1}}$ from Example 3.2,

$$\theta(V_\xi)(g, \mu) = \langle \mu, \text{Ad}_{g^{-1}} \xi \rangle = \langle \text{Ad}_g^* \mu, \xi \rangle = t^* \ell_\xi(g, \mu),$$

so $V_\xi = X_{t^* \ell_\xi}$.

It remains to differentiate along the flow $\tau \mapsto (\exp(\tau\xi)g, \mu)$ of V_ξ . Using $\frac{d}{d\tau} \Big|_0 \text{Ad}_{\exp(-\tau\xi)} = -\text{ad}_\xi$,

$$\begin{aligned} \{t^* \ell_\xi, t^* \ell_\eta\}_\Omega(g, \mu) &= \frac{d}{d\tau} \Big|_0 t^* \ell_\xi(\exp(\tau\xi)g, \mu) = \frac{d}{d\tau} \Big|_0 \langle \mu, \text{Ad}_{g^{-1}} \text{Ad}_{\exp(-\tau\xi)} \xi \rangle \\ &= \langle \mu, \text{Ad}_{g^{-1}} [\xi, \eta] \rangle = t^* \ell_{[\xi, \eta]}(g, \mu). \end{aligned}$$

By (3.3.2) the induced bracket satisfies $\{\ell_\xi, \ell_\eta\} = \ell_{[\xi, \eta]}$, which is the bracket of Definition 2.4. The source map is a Poisson map onto its negative, the bracket adopted in Remark 2.3 for the rigid body.

By Example 3.2 the source of an arrow is the covector recorded at the identity by left translation and the target is the same covector recorded by right translation. By Lemma 3.2(iii) a function of the first Poisson-commutes with a function of the second, and by Corollary 3.1(ii) the flow of a Hamiltonian pulled back along s does not move the target at all. This is Noether’s theorem for the free rigid body [8].

The source fibre over μ is $G \times \{\mu\}$, so for connected G the last two parts of Proposition 3.3 apply. The orbit of μ is $t(s^{-1}(\mu)) = \{\text{Ad}_g^* \mu : g \in G\}$, the coadjoint orbit through μ , and it is connected because G is. So the symplectic leaves of $\mathfrak{g}^*$ are the coadjoint orbits, which is the leaf statement of Theorem 2.2 obtained again from the groupoid. A function C is a Casimir exactly when $C(\mu) = C(\text{Ad}_g^* \mu)$ for every g , that is exactly when C is Ad^* -invariant, as Section 2.3.3 found. For $G = \text{SO}(3)$ the leaves are the spheres and $C(\mu) = \frac{1}{2}|\mu|^2$ is a Casimir. Section 3.5 recovers the bracket a second time from the Lie algebroid, without using θ .

3.4. Bisections

The algebroid of Chapter 1 carries infinitesimal motions of the base. Here we introduce the finite ones. A *bisection* is a choice of one arrow out of each point of the base. It moves

the base by a diffeomorphism, and when it is Lagrangian it moves the base by a Poisson diffeomorphism. Bisections of Lie groupoids are treated by Mackenzie [1]. Their Lagrangian version on a symplectic groupoid is due to Coste, Dazord and Weinstein [10].

Definition 3.3. A *bisection* of a Lie groupoid $\mathcal{G} \rightrightarrows M$ is a submanifold $B \subseteq \mathcal{G}$ on which both the source and the target restrict to diffeomorphisms onto M . A bisection of a symplectic groupoid is a *Lagrangian bisection* if in addition B is a Lagrangian submanifold of $(\mathcal{G}, \Omega)$.

A bisection meets each source fibre once and each target fibre once. The unit section $u(M)$ is the bisection of identity arrows. Every bisection induces a diffeomorphism of the base,

$$\psi_B = t|_B \circ (s|_B)^{-1}: M \longrightarrow M, \quad (3.4.1)$$

which carries a point x to the target of the arrow of B whose source is x .

Lemma 3.3. Let B be a bisection and put $\sigma = (s|_B)^{-1}: M \rightarrow \mathcal{G}$. Then σ is a smooth section of s with image B , and $t \circ \sigma = \psi_B$ is a diffeomorphism. Conversely, let $\sigma: M \rightarrow \mathcal{G}$ be a smooth section of s for which $t \circ \sigma$ is a diffeomorphism. Then $\sigma(M)$ is a bisection.

Proof. The first part is a restatement of the definition together with (3.4.1). For the converse, a section of a surjective submersion is an embedding, so $B = \sigma(M)$ is a submanifold and $s|_B$ is the inverse of σ , hence a diffeomorphism onto M . Then $t|_B$ factors as $(t \circ \sigma) \circ s|_B$, a composite of two diffeomorphisms. $\square$

We read both running examples through the lemma.

Example 3.5 (Bisections of the pair groupoid). Take $(M \times M, \Omega) \rightrightarrows M$ of Example 3.1, with $s(y, x) = x$ and $t(y, x) = y$. A section of s is a map $x \mapsto (\Phi(x), x)$ for a smooth $\Phi: M \rightarrow M$, and t composed with it is Φ itself. By Lemma 3.3 the bisections are exactly the graphs

$$\text{Gr}(\Phi) = \{(\Phi(x), x) : x \in M\}$$

of diffeomorphisms Φ of M , with $\psi_{\text{Gr}(\Phi)} = \Phi$. The bisection group is $\text{Diff}(M)$.

Such a graph is Lagrangian exactly when Φ is a symplectomorphism. Parametrise it by $\sigma(x) = (\Phi(x), x)$ and compute

$$\sigma^* \Omega = (\text{pr}_1 \circ \sigma)^* \omega - (\text{pr}_2 \circ \sigma)^* \omega = \Phi^* \omega - \omega.$$

The graph has dimension $\dim M$, which is half of $\dim(M \times M)$. So it is Lagrangian if and only if $\sigma^* \Omega = 0$. That is if and only if $\Phi^* \omega = \omega$. The Lagrangian bisections of the pair groupoid are the canonical transformations of (M, ω) .

Example 3.6 (Bisections of the cotangent groupoid). Take $T^*G \rightrightarrows \mathfrak{g}^*$ of Example 3.2, with $s(g, \mu) = \mu$ and $t(g, \mu) = \text{Ad}_g^* \mu$. A section of s is a map $\mu \mapsto (b(\mu), \mu)$ for a smooth $b: \mathfrak{g}^* \rightarrow G$. So the sections of the source are just the smooth maps into the group. By Lemma 3.3 such a section is a bisection exactly when

$$\psi_B(\mu) = \text{Ad}_{b(\mu)}^* \mu \quad (3.4.2)$$

is a diffeomorphism of $\mathfrak{g}^*$. Any b gives a section of the source, and the whole content of the definition sits in (3.4.2). It moves each μ inside its own coadjoint orbit. So ψ_B preserves every symplectic leaf and every Casimir of $\mathfrak{g}^*$, whatever b is. Section 4.3 is built on this.

Proposition 3.4. The bisections of $\mathcal{G}$ form a group under the product

$$B \cdot B' = \{bb' : b \in B, b' \in B', s(b) = t(b')\},$$

with unit $u(M)$ and with inverse $\iota(B)$. The map $B \mapsto \psi_B$ is a homomorphism into $\text{Diff}(M)$.

Proof. Let σ and σ' be the sections of Lemma 3.3. For $x \in M$ the arrow $\sigma'(x)$ has target $\psi_{B'}(x)$, and $\sigma(\psi_{B'}(x))$ has source $\psi_{B'}(x)$. So the two are composable and

$$\tau(x) = \sigma(\psi_{B'}(x)) \sigma'(x)$$

is defined and smooth. Its source is x , so τ is a section of s with image $B \cdot B'$. Its target is $\psi_B(\psi_{B'}(x))$. That composite is a diffeomorphism. So $B \cdot B'$ is a bisection by Lemma 3.3, and $\psi_{B \cdot B'} = \psi_B \circ \psi_{B'}$.

The unit section is a bisection with $\psi_{u(M)} = \text{id}_M$. Relations $b u(s(b)) = b$ and $u(t(b)) b = b$ make it a unit for the product. For the inverse, $s(b^{-1}) = t(b)$ and $t(b^{-1}) = s(b)$. So s and t restricted to $\iota(B)$ are $t|_B$ and $s|_B$ read through the inversion. Both are diffeomorphisms, so $\iota(B)$ is a bisection, and $\psi_{\iota(B)} = s|_B \circ (t|_B)^{-1} = \psi_B^{-1}$. Finally take $b, b' \in B$ with $s(b) = t((b')^{-1}) = s(b')$. Injectivity of $s|_B$ forces $b = b'$, so $B \cdot \iota(B) = \{b b^{-1}\} = u(M)$. The same argument with $t|_B$ gives $\iota(B) \cdot B = u(M)$. $\square$

A bisection also acts on the groupoid itself. This is how it reaches the base.

Proposition 3.5 (Left translation). *Let B be a bisection with section σ . The map*

$$A_B: \mathcal{G} \longrightarrow \mathcal{G}, \quad A_B(g) = \sigma(t(g)) g,$$

is a diffeomorphism of $\mathcal{G}$ with inverse $A_{\iota(B)}$, and it satisfies

$$s \circ A_B = s, \quad t \circ A_B = \psi_B \circ t.$$

If $\mathcal{G}$ is symplectic and B is Lagrangian, then A_B is a symplectomorphism of $(\mathcal{G}, \Omega)$.

Proof. The product is defined because $\sigma(t(g))$ has source $t(g)$. It is smooth as a composite of smooth maps. That $A_{\iota(B)}$ inverts A_B follows from Proposition 3.4. Multiplication does not change the source of the right factor or the target of the left factor, so $s(A_B(g)) = s(g)$ and $t(A_B(g)) = t(\sigma(t(g))) = \psi_B(t(g))$.

For the last claim write $A_B = m \circ \rho$, where $\rho(g) = (\sigma(t(g)), g)$ is a map into $\mathcal{G}^{(2)}$. Multiplicativity (3.1.1) gives

$$A_B^* \Omega = \rho^* m^* \Omega = (\text{pr}_1 \circ \rho)^* \Omega + (\text{pr}_2 \circ \rho)^* \Omega = (\sigma \circ t)^* \Omega + \Omega.$$

Now $(\sigma \circ t)^* \Omega = t^* \sigma^* \Omega$, and $\sigma^* \Omega = 0$ because σ parametrises B and B is Lagrangian. So $A_B^* \Omega = \Omega$. $\square$

Proposition 3.6. *Let $(\mathcal{G} \rightrightarrows M, \Omega)$ be a symplectic groupoid and give M the Poisson structure π of Theorem 3.1. If B is a Lagrangian bisection then ψ_B is a Poisson diffeomorphism of (M, π) . The Lagrangian bisections form a subgroup of the bisection group, and $B \mapsto \psi_B$ restricts to a homomorphism of that subgroup into the Poisson diffeomorphisms of (M, π) .*

Proof. By Proposition 3.5 the map A_B is a symplectomorphism of $(\mathcal{G}, \Omega)$, hence a Poisson map for the bracket of Ω . It covers ψ_B through the target, which is Poisson by Theorem 3.1. Thus the composite $\psi_B \circ t = t \circ A_B$ is Poisson. The target is also a surjective submersion, so t^* is injective. This forces ψ_B itself to be Poisson.

For the subgroup, note that $B \cdot B' = A_B(B')$. If B and B' are Lagrangian then A_B is a symplectomorphism, so $A_B(B')$ is Lagrangian. The inversion is anti-symplectic by Lemma 3.1(iii). A form and its negative have the same Lagrangian submanifolds, so $\iota(B)$ is Lagrangian too. The homomorphism property is Proposition 3.4. $\square$

A bisection does not need to be Lagrangian to respect the degeneracy of π . By (3.4.1) the points x and $\psi_B(x)$ are the two ends of one arrow, so they lie in one orbit of $\mathcal{G}$. Let the source fibres of $\mathcal{G}$ be connected. Then ψ_B preserves each symplectic leaf by Proposition 3.3(ii), and $C \circ \psi_B = C$ for every Casimir C by part (iii) of the same proposition. Example 3.6 shows this for the cotangent groupoid.

Remark 3.3. The converse fails in general. On an arbitrary symplectic groupoid not every Poisson diffeomorphism of the base is induced by a Lagrangian bisection. On the pair groupoid every bisection is a graph and the computation of Example 3.5 is an equivalence. So over a symplectic base the two notions agree, and the canonical transformations of classical mechanics are exactly the Lagrangian bisections. Over a degenerate base the correspondence runs only one way. The substitute for a Hamiltonian flow is the following proposition.

Proposition 3.7. *Let $(\mathcal{G} \rightrightarrows M, \Omega)$ be a symplectic groupoid, let π be the induced structure, let $H \in C^\infty(M)$ have Hamiltonian flow φ_τ on (M, π) , and let $\widehat{\varphi}_\tau$ be the flow on $\mathcal{G}$ of the Hamiltonian vector field X_{s^*H} . For every τ for which $\widehat{\varphi}_\tau$ is defined on all of $u(M)$, the set*

$$B_\tau := \widehat{\varphi}_\tau(u(M))$$

is a Lagrangian bisection of $\mathcal{G}$, with $B_0 = u(M)$ and $\psi_{B_\tau} = \varphi_\tau$.

Proof. B_τ is Lagrangian. The map $\widehat{\varphi}_\tau$ is the flow of a Hamiltonian vector field on $(\mathcal{G}, \Omega)$, so it is a symplectomorphism where it is defined. The unit section is Lagrangian by Proposition 3.2, and a symplectomorphism carries Lagrangian submanifolds to Lagrangian submanifolds.

B_τ is a bisection with base map φ_τ . By Corollary 3.1(ii) the flow satisfies $t \circ \widehat{\varphi}_\tau = t$ and $s \circ \widehat{\varphi}_\tau = \varphi_{-\tau} \circ s$. Hence $t \circ (\widehat{\varphi}_\tau \circ u) = t \circ u = \text{id}_M$ and $s \circ (\widehat{\varphi}_\tau \circ u) = \varphi_{-\tau}$. Now put $\sigma_\tau = (\widehat{\varphi}_\tau \circ u) \circ \varphi_\tau$. Then $s \circ \sigma_\tau = \varphi_{-\tau} \circ \varphi_\tau = \text{id}_M$, so σ_τ is a smooth section of s , and $t \circ \sigma_\tau = \varphi_\tau$ is a diffeomorphism. By Lemma 3.3 its image is a bisection with base map φ_τ . That image is B_τ , because φ_τ is a bijection of M . $\square$

For the pair groupoid the proposition returns $B_\tau = \text{Gr}(\varphi_\tau)$, so a Hamiltonian flow is a one-parameter subgroup of the group of canonical transformations. For the cotangent groupoid it is equally concrete. By Example 3.6 the bisection B_τ is a smooth map $b_\tau: \mathfrak{g}^* \rightarrow G$ with $b_0 \equiv e$, and $\psi_{B_\tau} = \varphi_\tau$ reads

$$\text{Ad}_{b_\tau(\mu)}^* \mu = \varphi_\tau(\mu). \quad (3.4.3)$$

The exact flow of any Hamiltonian on $\mathfrak{g}^*$ therefore moves each momentum by a coadjoint transformation, with the group element chosen smoothly. Conservation of every Casimir follows from the shape of (4.2.2) and from nothing about H .

Remark 3.4 (Generating functions). Close to the unit section the Lagrangian bisections are described by functions on M . The unit section is Lagrangian by Proposition 3.2 and diffeomorphic to M , so the Lagrangian neighbourhood theorem makes some neighbourhood of $u(M)$ in $(\mathcal{G}, \Omega)$ symplectomorphic to a neighbourhood of the zero section in T^*M , with $u(M)$ going to the zero section [5]. A Lagrangian submanifold of T^*M that is a section of the projection is the graph of a closed one-form, and locally a closed one-form is dS for a function S on M . So one function encodes one Lagrangian bisection near the units. Such an S is called a *generating function*. Section 4.3 builds numerical methods from generating functions on the cotangent groupoid, where no neighbourhood theorem is needed, because T^*G is itself a cotangent bundle and the graph of dS for a function S on the group is Lagrangian from the start [18, 19].

3.5. The Lie algebroid of a symplectic groupoid

Chapter 1 defines a Lie algebroid of a Lie groupoid. Chapter 2 defines the cotangent Lie algebroid of a Poisson manifold (Theorem 2.3). Theorem 3.1 shows that every symplectic groupoid admits Poisson structure on its base. This section shows that the three constructions close on each other. The Lie algebroid of a symplectic groupoid is the cotangent algebroid

of its base. The result is due to Coste, Dazord and Weinstein [10]. In the lectures of Crainic and Fernandes it is stated as one theorem together with Theorem 3.1 [15].

The identification rests on linear algebra at the units. Fix $x \in M$. The fibre $A(\mathcal{G})_x = \ker T_{1_x}t$ and the subspace $T_{1_x}u(M) = Tu(T_xM)$ are transverse in $T_{1_x}\mathcal{G}$. That is because a vector in both is $Tu(a)$ with $a = Tt(Tu(a)) = 0$, and the dimensions add to $\dim M + \dim M = \dim \mathcal{G}$ by Proposition 3.2. Now suppose $v \in A(\mathcal{G})_x$ satisfies $\Omega_{1_x}(v, Tu(a)) = 0$ for every $a \in T_xM$. That means that v lies in the symplectic orthogonal of $Tu(T_xM)$. The subspace $Tu(T_xM)$ is Lagrangian by the same proposition, and a Lagrangian subspace equals its own symplectic orthogonal. So v lies both in $Tu(T_xM)$ and in $A(\mathcal{G})_x$. The transversality gives $v = 0$. The following pairing is therefore injective, and the two spaces have equal dimension. Hence

$$v \longmapsto -\Omega_{1_x}(v, Tu(\cdot))$$

is a linear isomorphism from $A(\mathcal{G})_x$ onto T_x^*M . The sign is a choice, fixed so that the anchors match in the theorem.

Theorem 3.2 (Coste–Dazord–Weinstein). *Let $(\mathcal{G} \rightrightarrows M, \Omega)$ be a symplectic groupoid and let π be the induced structure on M . The bundle map*

$$\Psi: A(\mathcal{G}) \longrightarrow T^*M, \quad \langle \Psi(v), a \rangle = -\Omega_{1_x}(v, Tu(a)) \quad (v \in A(\mathcal{G})_x, a \in T_xM),$$

is an isomorphism of Lie algebroids onto the cotangent algebroid of (M, π) . It intertwines the anchors, $\pi^\sharp \circ \Psi = \rho$, it carries the bracket of $A(\mathcal{G})$ to the Koszul bracket, and it satisfies

$$\Psi(X_{s^*f_1}(1_x)) = -d(f_1)_x, \quad f_1 \in C^\infty(M). \quad (3.5.1)$$

Proof. Ψ is an isomorphism of vector bundles. The paragraph before the theorem gives the fibrewise isomorphism, and Ψ is smooth because Ω and Tu are.

The identity (3.5.1). The vector $X_{s^*f_1}(1_x)$ lies in $\ker T_{1_x}t = A(\mathcal{G})_x$ by Lemma 3.2(i). For $a \in T_xM$,

$$\langle \Psi(X_{s^*f_1}(1_x)), a \rangle = -\Omega_{1_x}(X_{s^*f_1}(1_x), Tu(a)) = -d(s^*f_1)(Tu(a)) = -df_1(TsTu(a)) = -df_1(a),$$

because $\iota_{X_{s^*f_1}}\Omega = d(s^*f_1)$ and $s \circ u = \text{id}_M$.

The spanning sections. For $f_1 \in C^\infty(M)$ put

$$\alpha_{f_1} = \Psi^{-1}(df_1) = -X_{s^*f_1}|_{u(M)} \in \Gamma(A(\mathcal{G})).$$

The differentials $d(f_1)_x$ fill T_x^*M as f_1 varies, so the values $\alpha_{f_1}(x)$ fill $A(\mathcal{G})_x$. The field $-X_{s^*f_1}$ takes values in $\ker Tt$ and is left-invariant, by Lemma 3.2(i) and (ii). A left-invariant field is determined by its values at the units (Section 1.3.1). So the left-invariant extension of α_{f_1} is

$$\overleftarrow{\alpha}_{f_1} = -X_{s^*f_1}.$$

The anchors. By Corollary 3.1(ii) and by $X_{f_1} = \pi^\sharp df_1$ (Section 2.2.1),

$$\rho(\alpha_{f_1})(x) = Ts(-X_{s^*f_1}(1_x)) = X_{f_1}(x) = \pi^\sharp(df_1)_x = \pi^\sharp(\Psi\alpha_{f_1})(x).$$

The values $\alpha_{f_1}(x)$ fill every fibre, ρ and $\pi^\sharp \circ \Psi$ are bundle maps, so $\rho = \pi^\sharp \circ \Psi$.

The brackets on the spanning sections. The bracket of $A(\mathcal{G})$ restricts the bracket of left-invariant fields to the units (Section 1.3.1), so

$$[\alpha_{f_1}, \alpha_{f_2}] = [X_{s^*f_1}, X_{s^*f_2}]|_{u(M)} = -X_{\{s^*f_1, s^*f_2\}_\Omega}|_{u(M)} = X_{s^*\{f_1, f_2\}}|_{u(M)} = -\alpha_{\{f_1, f_2\}},$$

where the second step is Proposition 2.1(ii) on $(\mathcal{G}, \Omega)$ and the third is the anti-Poisson property of s from Theorem 3.1. Applying Ψ and using (2.4.1),

$$\Psi([\alpha_{f_1}, \alpha_{f_2}]) = -d\{f_1, f_2\} = [df_1, df_2]_\pi = [\Psi\alpha_{f_1}, \Psi\alpha_{f_2}]_\pi.$$

The brackets on all sections. Transport the Koszul bracket through Ψ , that is put $[\alpha, \beta]' = \Psi^{-1}[\Psi\alpha, \Psi\beta]_\pi$. It is antisymmetric and satisfies the Jacobi identity because the

Koszul bracket does. It satisfies the Leibniz rule with the anchor ρ , since by Theorem 2.3 and by the anchor identity

$$[\alpha, f_1\beta]' = \Psi^{-1}\left(f_1[\Psi\alpha, \Psi\beta]_\pi + (\pi^\sharp\Psi\alpha)(f_1)\Psi\beta\right) = f_1[\alpha, \beta]' + \rho(\alpha)(f_1)\beta.$$

The brackets $[\cdot, \cdot]$ and $[\cdot, \cdot]'$ agree on the sections α_{f_1} . Around any point choose $c_1, \dots, c_n \in C^\infty(M)$ so that $dc_1, \dots, dc_n$ frame T^*M . Then $\alpha_{c_1}, \dots, \alpha_{c_n}$ frame $A(\mathcal{G})$, and every section expands in this frame with smooth coefficients. Both brackets are antisymmetric and expand through the same Leibniz rule with the same anchor. In the expansions the coefficients multiply brackets of the α_{c_i} , which agree, and the derivative terms contain only the anchor. So the two brackets agree on all sections, and Ψ carries $[\cdot, \cdot]$ to $[\cdot, \cdot]_\pi$. $\square$

Remark 3.5 (Conventions). Sources differ over the algebroid of a groupoid as well. Section 1.3.1 builds $A(\mathcal{G})$ from $\ker(Tt)$ with left-invariant fields and anchor Ts . Other references build it from $\ker(Ts)$ with right-invariant fields and anchor Tt . Together with the orientation of Remark 3.2 this changes signs in Theorem 3.2. A reader comparing with the literature should check the two conventions together.

Two readings of the theorem are used later. First, $A(\mathcal{G})_x$ is a complement of $T_{1_x}u(M)$, so the quotient map identifies $A(\mathcal{G})$ with the normal bundle of the unit section, and Ψ then identifies that normal bundle with T^*M . This is the bundle in which Remark 3.4 writes generating functions. Second, the fibre $A(\mathcal{G})_x$ is not isotropic in general. On it the symplectic form computes the bracket of the base, since

$$\Omega(X_{s^*f_1}, X_{s^*f_2})(1_x) = \{s^*f_1, s^*f_2\}_\Omega(1_x) = -\{f_1, f_2\}(x),$$

by the anti-Poisson property of s and $s \circ u = \text{id}_M$. So $A(\mathcal{G})_x$ is isotropic exactly at the zeros of π .

The invariants of the two algebroids also match. Chapter 1 states that the orbits and the isotropy Lie algebras of $A(\mathcal{G})$ are those of $\mathcal{G}$. Section 2.4.2 states that the orbits of the cotangent algebroid are the symplectic leaves and that its isotropy at x is $\ker \pi_x^\sharp$. Under Ψ the two statements agree, and they agree with Proposition 3.3(ii). In particular the Lie algebra of the vertex group $\mathcal{G}_x^x$ is carried by Ψ onto $\ker \pi_x^\sharp$, the conormal space of the leaf with its transverse bracket.

Example 3.7 (The algebroid of the pair groupoid). Take $(M \times M, \Omega) \rightrightarrows M$ over a symplectic manifold (M, ω) , with $t = \text{pr}_1$, $s = \text{pr}_2$ and $\Omega = \text{pr}_1^*\omega - \text{pr}_2^*\omega$. Chapter 1 computes its Lie algebroid in Example 1.5. The fibre at the unit (x, x) is $\ker T \text{pr}_1 = \{(0, b) : b \in T_x M\}$ identified with $T_x M$, and the anchor is the identity. The induced structure on the base is the bracket of ω by Example 3.3. With $Tu(a) = (a, a)$,

$$\langle \Psi(0, b), a \rangle = -\Omega((0, b), (a, a)) = -(\omega(0, a) - \omega(b, a)) = \omega(b, a),$$

so $\Psi(b) = \omega^\flat(b)$. The theorem says that $\omega^\flat$ is an isomorphism from the tangent algebroid onto the cotangent algebroid of (M, ω) . That is the isomorphism of Example 2.3 read in the opposite direction.

Example 3.8 (The algebroid of the cotangent groupoid). Take $T^*G \rightrightarrows \mathfrak{g}^*$ of Example 3.2, with $s(g, \mu) = \mu$, $t(g, \mu) = \text{Ad}_g^*\mu$ and $(h, \text{Ad}_g^*\mu)(g, \mu) = (hg, \mu)$. These are the structure maps of the action groupoid $G \ltimes \mathfrak{g}^*$ of the coadjoint action (Example 1.3). So Example 1.6 computes the Lie algebroid. It is the action algebroid $\mathfrak{g} \ltimes \mathfrak{g}^*$, the element $\xi \in \mathfrak{g}$ sits at the unit over μ as the kernel vector

$$w_\xi = (\xi, \text{ad}_\xi^*\mu) \in \ker T_{(e, \mu)}t,$$

the anchor sends it to $\text{ad}_\xi^*\mu$, and constant sections bracket to the bracket of $\mathfrak{g}$.

To apply (3.5.1) we need $X_{s^*\ell_\xi}$ at the units. The computation mirrors the one for Λ_h in Example 3.4. For $h \in G$ put $P_h(g, \nu) = (gh, \text{Ad}_{h^{-1}}^* \nu)$. Using $C_{h^{-1}} = L_{h^{-1}} \circ R_h$ and $T_e C_{h^{-1}} = \text{Ad}_{h^{-1}}$ from Example 3.2,

$$\theta_{gh}^L \circ T_g R_h = T(L_{(gh)^{-1}} \circ R_h) = T(C_{h^{-1}} \circ L_{g^{-1}}) = \text{Ad}_{h^{-1}} \circ \theta_g^L,$$

and therefore

$$\langle \text{Ad}_{h^{-1}}^* \nu, \theta_{gh}^L(T_g R_h v) \rangle = \langle \text{Ad}_{h^{-1}}^* \nu, \text{Ad}_{h^{-1}}(\theta_g^L v) \rangle = \langle \nu, \theta_g^L v \rangle,$$

so $P_h^* \theta = \theta$. Let W_ξ generate $\tau \mapsto P_{\exp(\tau\xi)}$. By Section 2.3.3 the derivative of $\tau \mapsto \text{Ad}_{\exp(-\tau\xi)}^* \nu$ at zero is $\text{ad}_\xi^* \nu$, so

$$W_\xi(g, \nu) = (T_e L_g \xi, \text{ad}_\xi^* \nu).$$

Cartan's formula with $\Omega = -d\theta$ gives $\iota_{W_\xi} \Omega = d(\theta(W_\xi))$, and

$$\theta(W_\xi)(g, \nu) = \langle \nu, \theta_g^L(T_e L_g \xi) \rangle = \langle \nu, \xi \rangle = s^* \ell_\xi(g, \nu),$$

so $W_\xi = X_{s^*\ell_\xi}$. At the unit over μ its value is $W_\xi(e, \mu) = w_\xi$, the constant section ξ . By (3.5.1),

$$\Psi(w_\xi) = -d\ell_\xi, \quad \text{that is} \quad \Psi(\mu, \xi) = (\mu, -\xi).$$

This is the map Φ of Example 2.4 read in the opposite direction, with the same formula. The minus sign that Chapter 2 inserted by hand to match the anchors is produced here by the symplectic form. The suggestion at the end of Example 2.4 is now confirmed twice. The groupoid T^*G induces the Lie–Poisson structure on its base by Example 3.4, and its Lie algebroid is the cotangent algebroid of that structure.

3.6. Integrability of Poisson manifolds

Theorems 3.1 and 3.2 pass from a symplectic groupoid to a Poisson manifold and identify the Lie algebroid of the groupoid with its cotangent algebroid. This section concerns the converse question: given a Poisson manifold (M, π) , does a symplectic groupoid over it exist? Here the analogy with Lie theory breaks down. Every finite-dimensional Lie algebra is the Lie algebra of a Lie group, but a Poisson manifold need not be integrable. We state the integrability criterion, exhibit a Poisson manifold for which it fails, and record that the integration, when it exists, carries a canonical symplectic form; the proofs are in the cited literature.

The failure is global, never local. Near any point a Poisson manifold splits (Theorem 2.1) into a symplectic factor, integrated by its pair groupoid, and a transverse Poisson factor whose rank vanishes at the point. Over a neighbourhood of each point one can construct a *local symplectic groupoid*: a symplectic manifold whose multiplication is defined and associative for arrows near the units, satisfies the groupoid axioms where defined, and induces π on the base. Such a local object always exists and is unique near the units. Weinstein and, independently, Karasev constructed it from a *generating function*, a solution of a Hamilton–Jacobi equation determined by π , whose graph realises the Lagrangian multiplication of Remark 3.1 near the units [9, 11, 10]; the same generating function is used in Section 4.3. The obstruction lies in patching the local groupoids over the whole of M into a single Lie groupoid with globally defined, associative multiplication.

When the patching succeeds there is a canonical integration, constructed as for general Lie algebroids (Remark 1.3): the *Weinstein groupoid* $\Sigma(M, \pi) \rightrightarrows M$, whose arrows are suitable paths in T^*M taken modulo homotopy and composed by concatenation, in analogy with the construction of the simply connected Lie group from paths in $\mathfrak{g}$. The groupoid $\Sigma(M, \pi)$ is always defined as a topological groupoid. When it is a smooth manifold, it is the unique *source-simply-connected* symplectic groupoid integrating (M, π) , in particular with

connected target fibres as assumed in Theorem 3.1, and every other integration is a quotient of it. Smoothness is governed by a computable invariant of the symplectic leaves, the *monodromy*, which measures how the symplectic areas of neighbouring leaves vary in the directions transverse to a leaf; it is a π_2 effect, with no counterpart for a Lie algebra, whose leaves are points.

Theorem 3.3 (Crainic–Fernandes). *A Poisson manifold (M, π) is integrable by a symplectic groupoid if and only if the monodromy groups of its symplectic leaves are uniformly discrete, in the sense made precise in [14]. When this holds, the source-simply-connected groupoid $\Sigma(M, \pi)$ is a smooth symplectic groupoid integrating (M, π) , of which every integration is a quotient; when it fails, the source fibres of $\Sigma(M, \pi)$ are singular and no symplectic groupoid over (M, π) exists.*

The obstruction is realised by the following rotationally invariant deformation of the Lie–Poisson structure of $\mathfrak{so}(3)^*$.

Example 3.9 (A Poisson manifold with no symplectic groupoid). On $\mathbb{R}^3 \setminus \{0\}$ write $r = |x|$ and take

$$\{x_i, x_j\} = f(r) \sum_k \varepsilon_{ijk} x_k,$$

with $f > 0$ a smooth function of the radius alone. For $f \equiv 1$ this is the Lie–Poisson structure of $\mathfrak{so}(3)^*$ (Section 2.3), punctured at the origin and integrated by the cotangent groupoid $T^*\mathrm{SU}(2)$ (Example 3.2). For any f the symplectic leaves are unchanged, the spheres $r = R$; only the leafwise area is rescaled, to

$$A(R) = \frac{4\pi R}{f(R)}.$$

Each leaf is a two-sphere and contributes a single period, its area, so the monodromy at radius R lies in the radial line transverse to the leaf and is generated by the variation of A , that is by $A'(R)$. Where A is strictly monotone these groups are uniformly discrete and the structure is integrable. If f is chosen so that A has a critical point at some R_0 , with $A'(R_0) = 0$ and $A' \neq 0$ on either side, then the generators $A'(R)$ tend to zero as $R \rightarrow R_0$, the discreteness criterion of Theorem 3.3 fails, the would-be source fibres are singular, and no symplectic groupoid exists over any neighbourhood of the sphere $r = R_0$. The example is Weinstein’s, through a theorem of Dazord; the area computation is due to Cattaneo and Felder [16]. For the linear structure $A(R) = 4\pi R$ has no critical point, which is why the obstruction does not appear there.

The Poisson manifolds appearing in this thesis are integrable. A symplectic manifold is integrated by its pair groupoid (Example 3.1); strictly, the source-simply-connected integration is the fundamental groupoid, which coincides with $M \times M$ when M is simply connected. The dual $\mathfrak{g}^*$ is integrated by T^*G (Example 3.2), the case $f \equiv 1$ of Example 3.9. When the integration exists, the smooth groupoid $\Sigma(M, \pi)$ carries a canonical multiplicative symplectic form; the form is determined by π and unique.

Chapter 4

Physical Applications

The previous chapter established, in the abstract, that symplectic groupoids integrate Poisson manifolds. This chapter shows what the abstraction does for mechanics. It moves in three steps. First the groupoid is a *language* for dynamics one already knows. Then it is the carrier of a *theorem* one already needs. Finally it is a *tool* that produces something the Poisson manifold cannot give on its own.

Section 4.1 treats the non-degenerate case, the pair groupoid of a symplectic phase space. Its Lagrangian bisections are the canonical transformations, and Hamiltonian flows are paths of bisections. This short dictionary is used by everything that follows. Section 4.2 treats the degenerate case, the cotangent groupoid $T^*G \rightrightarrows \mathfrak{g}^*$ and the free rigid body. The main point is that Lie–Poisson reduction is not merely organised by the groupoid. It *is* Theorem 3.1, specialised to T^*G . Section 4.3 is the payoff: numerical methods that respect the Poisson geometry. No background in numerical analysis is assumed there. The few notions needed are introduced from scratch, and the section is built as a sequence of small ideas, each stated together with what it buys.

A remark on authorship is in order. Almost every result below is classical or belongs to recent published work, and we cite it as such. The contribution of these notes is expository. We show that the structure theory of Chapter 3, proved there once for all symplectic groupoids, already contains the reduction theory of the rigid body and explains the design of modern Poisson integrators. Facts quoted from numerical analysis carry a reference to the book of Hairer, Lubich and Wanner [17]. The quantisation programme that historically motivated symplectic groupoids lies beyond these notes; the Conclusion records where it would begin.

4.1. The pair groupoid: canonical transformations and dynamics

We begin with the non-degenerate case, where the base is itself symplectic. Here the groupoid adds nothing that the symplectic form did not already contain, and its value is a precise vocabulary.

Let (M, ω) be a symplectic phase space. Its symplectic groupoid is the pair groupoid $(M \times M, \Omega) \rightrightarrows M$, with source $s(y, x) = x$, target $t(y, x) = y$, multiplication $(z, y)(y, x) = (z, x)$ and the multiplicative form $\Omega = \text{pr}_1^* \omega - \text{pr}_2^* \omega$ (Example 3.1).

An arrow (y, x) records a transition from the phase point x to the phase point y . A deterministic rule of motion assigns to each starting point exactly one endpoint, and each endpoint is reached from exactly one starting point. As a set of arrows, such a rule meets every

source value once and every target value once. These sets are the bisections of Definition 3.3. The symplectic form now singles out, among all bisections, the ones that mechanics allows.

A diffeomorphism $\Phi: M \rightarrow M$ is *canonical* (a symplectomorphism) if $\Phi^*\omega = \omega$. Canonical transformations are the symmetries of Hamiltonian mechanics. They preserve the bracket, and so carry Hamiltonian systems to Hamiltonian systems. Inside the groupoid, a diffeomorphism appears as its graph,

$$\text{Gr}(\Phi) = \{(\Phi(x), x) : x \in M\} \subset M \times M,$$

the set of arrows implementing the rule $x \mapsto \Phi(x)$. The graph is a bisection, with induced base map $\psi_{\text{Gr}(\Phi)} = \Phi$, and every bisection of the pair groupoid is a graph. Which graphs are canonical is decided by a two-line computation.

Proposition 4.1. *A diffeomorphism Φ of (M, ω) is canonical if and only if $\text{Gr}(\Phi)$ is a Lagrangian bisection of $(M \times M, \Omega)$.*

Proof. Parametrise the graph by $i: M \rightarrow M \times M$, $i(x) = (\Phi(x), x)$. Then

$$i^*\Omega = (\text{pr}_1 \circ i)^*\omega - (\text{pr}_2 \circ i)^*\omega = \Phi^*\omega - \omega.$$

The graph has dimension $\dim M$, half the dimension of $M \times M$, so it is Lagrangian if and only if $i^*\Omega = 0$, that is, if and only if $\Phi^*\omega = \omega$. $\square$

Remark 4.1. It is worth comparing this with Proposition 3.6, and the comparison is not symmetric. One direction is a special case. If $\text{Gr}(\Phi)$ is a Lagrangian bisection, then Φ is a Poisson diffeomorphism of the base by Proposition 3.6. The base structure of the pair groupoid is the canonical bracket (Example 3.3), and preserving that bracket is the same as preserving ω (Example 2.1). The other direction is extra. Proposition 3.6 does not say that every Poisson diffeomorphism of the base comes from a Lagrangian bisection, and on a general symplectic groupoid this converse is a delicate question. On the pair groupoid it happens to hold, because every bisection is a graph and the computation above is an equivalence. For the maps we will actually need, the Hamiltonian flows, the general substitute for the missing converse is Proposition 4.3 in the next section.

Mechanics is not a single transformation but a one-parameter family of them. Let $H \in C^\infty(M)$ generate the flow φ_t , so that $\dot{f} = \{f, H\}$ along trajectories. Each φ_t is canonical, so the flow traces a path of Lagrangian bisections

$$B_t = \text{Gr}(\varphi_t), \quad B_0 = \Delta = u(M),$$

starting at the unit bisection. The path respects the groupoid structure. Indeed $B_{t+s} = B_t \cdot B_s$ in the product of bisections, because $\varphi_{t+s} = \varphi_t \circ \varphi_s$, so a Hamiltonian flow is a one-parameter subgroup of the group of Lagrangian bisections.

Differentiating the path at $t = 0$ gives a tangent vector at each unit. Modulo directions along the diagonal, that vector is $(X_H, 0)$, and under the identification $A(M \times M) \cong TM$ of Example 3.7 it is the Hamiltonian vector field X_H itself. The dictionary of this section is now complete. Finite motions are Lagrangian bisections, infinitesimal motions are sections of the algebroid, and the flow passes between the two. The degenerate case is next, and there the groupoid starts to earn its place.

4.2. The cotangent groupoid: reduction and the rigid body

The physical story of this section is symmetry reduction. A rigid body has a six-dimensional phase space T^*G , but its rotational symmetry cuts the description down to three dimensions, and the three-dimensional space is the Lie–Poisson manifold $\mathfrak{g}^*$ of Section 2.3. This reduction

is classical [8]. What the groupoid adds is the claim that Chapter 3 has already proved it. The section makes three points. The reduction is a short theorem whose entire content comes from the structure theory of Chapter 3. The conserved quantities of the theory, the momentum maps, are the structure maps s and t . And every Hamiltonian flow on $\mathfrak{g}^*$ is covered by a path of Lagrangian bisections, which is the statement that the numerical section will discretise.

The rigid body and its two frames

A rigid body fixed at its centre of mass has configuration space $G = \mathrm{SO}(3)$. The rotation $R \in G$ carries a frame glued to the body onto a frame fixed in space, and the phase space is T^*G . Every mechanical quantity can be reported in two frames. Watching from the laboratory one records the *spatial* angular momentum $\mathbf{m}$. Riding with the body one records the *body* angular momentum $\boldsymbol{\mu}$. The two are related by $\mathbf{m} = R\boldsymbol{\mu}$. (We reserve π for Poisson structures and write $\mathbf{m}$ for the spatial momentum.)

The free dynamics is simplest in the body frame. With inertia tensor $\mathbb{I} = \mathrm{diag}(I_1, I_2, I_3)$ in principal axes and body angular velocity $\boldsymbol{\Omega} = \mathbb{I}^{-1}\boldsymbol{\mu}$, Euler's equation reads

$$\dot{\boldsymbol{\mu}} = \boldsymbol{\mu} \times \boldsymbol{\Omega},$$

a flow on the three-dimensional space $\boldsymbol{\mu} \in \mathbb{R}^3 \cong \mathfrak{so}(3)^*$ rather than on the six-dimensional T^*G .

Example ?? already exhibited the target of this passage. It is $\mathfrak{so}(3)^*$ with a linear Poisson bracket, the spheres $|\boldsymbol{\mu}| = \mathrm{const}$ as symplectic leaves, and $\frac{1}{2}|\boldsymbol{\mu}|^2$ as a Casimir. What Chapter 2 could not say is where this bracket comes from, and why $\mathfrak{g}^*$ carries two natural signs $\pi_{\pm}$. Both questions are answered by the groupoid.

Recall from Example 3.2 the cotangent groupoid of a Lie group G . Trivialising $T^*G \cong G \times \mathfrak{g}^*$ by left translations, $\alpha_g \mapsto (g, \mu)$ with $\mu = (T_e L_g)^* \alpha_g$, the structure maps are

$$s(g, \mu) = \mu, \quad t(g, \mu) = \mathrm{Ad}_g^* \mu, \quad u(\mu) = (e, \mu), \quad \iota(g, \mu) = (g^{-1}, \mathrm{Ad}_g^* \mu),$$

with multiplication $(g, \mathrm{Ad}_h^* \mu)(h, \mu) = (gh, \mu)$, and the canonical symplectic form $\Omega = -d\theta$, $\theta = \langle \mu, \theta^L \rangle$, is multiplicative. The physical reading is direct. An arrow (g, μ) is a rotation g applied to a body state with momentum μ . Its source is the body momentum, and its target $\mathrm{Ad}_g^* \mu$ is the same momentum reported in the rotated frame. For $G = \mathrm{SO}(3)$ the coadjoint action is the ordinary action of rotations on vectors, so $t(R, \boldsymbol{\mu}) = R\boldsymbol{\mu} = \mathbf{m}$. In short, *the source is the body momentum and the target is the spatial momentum*. The two frames of rigid-body mechanics are the two structure maps of the groupoid.

Reduction of the phase space

The symmetry that reduction uses is the action of G on itself by left translations, lifted to T^*G . In the left trivialisation the lift takes one line to compute. The cotangent lift of L_h sends α_g to $(T_{hg} L_{h^{-1}})^* \alpha_g$, whose body coordinate is

$$(T_e L_{hg})^* (T_{hg} L_{h^{-1}})^* \alpha_g = (T_e (L_{h^{-1}} \circ L_{hg}))^* \alpha_g = (T_e L_g)^* \alpha_g = \mu,$$

so that

$$h \cdot (g, \mu) = (hg, \mu). \tag{4.2.1}$$

The action moves the rotation and leaves the body momentum alone, as it must, since a spatial rotation cannot be detected from inside the body.

Seen with groupoid eyes, the action (4.2.1) is itself made of bisections. The subset $B_h = \{h\} \times \mathfrak{g}^*$ is a bisection, and the product rule gives $B_h \cdot B_k = B_{hk}$, so $h \mapsto B_h$ embeds G into the group of bisections. Left translation by B_h in the sense of Definition 3.3 is exactly (4.2.1).

Each B_h is also *Lagrangian*. It has half dimension, and θ vanishes on it because tangent vectors to B_h have no G -component, so $\Omega|_{B_h} = -d(\theta|_{B_h}) = 0$. By the computation in the proof of Proposition 3.6, left translation by a Lagrangian bisection preserves Ω . So the symmetry acts by symplectomorphisms, and this is read off from the multiplicativity of Ω rather than checked by hand. The induced base maps $\psi_{B_h} = \text{Ad}_h^*$ are Poisson diffeomorphisms by the same proposition. This recovers the standard fact that the coadjoint action preserves the Lie–Poisson bracket [8], without a bracket computation.

One more observation and the theorem states itself. A function $F \in C^\infty(T^*G)$ is invariant under (4.2.1) if and only if $F(g, \mu) = F(e, \mu)$ for all g , that is, if and only if $F = s^*f$ for a unique $f \in C^\infty(\mathfrak{g}^*)$. The orbits of the action are the fibres of s , so the quotient space is $\mathfrak{g}^*$ and the quotient map is s . The following theorem is classical, and direct proofs on T^*G can be found in [8]. The proof below is the point of this section. Every step of it is a reference to Chapter 3.

Theorem 4.1 (Lie–Poisson reduction). *Identify $T^*G \cong G \times \mathfrak{g}^*$ by left translations and let G act by (4.2.1). Then:*

- (i) *the invariant functions are exactly the functions of the body momentum, $s^*C^\infty(\mathfrak{g}^*)$, and the canonical bracket of two invariant functions is again invariant, so the bracket descends to the quotient $T^*G/G \cong \mathfrak{g}^*$;*
- (ii) *the descended bracket is minus the Lie–Poisson bracket of Definition 2.4,*

$$\{s^*f, s^*g\}_\Omega = s^*\{f, g\}_-, \quad \{f, g\}_-(\mu) = -\langle \mu, [d_\mu f, d_\mu g] \rangle,$$

*and the canonical dynamics of an invariant Hamiltonian s^*h descends to the Lie–Poisson dynamics of h on $(\mathfrak{g}^*, \pi_-)$. For the free rigid body this is Euler’s equation $\dot{\boldsymbol{\mu}} = \boldsymbol{\mu} \times \boldsymbol{\Omega}$ in the body frame.*

Proof. (i) The invariants and the quotient were identified above. The action preserves Ω , so the bracket of two invariant functions is invariant, and the bijection $f \leftrightarrow s^*f$ transports the bracket, with its Jacobi identity and Leibniz rule, to $C^\infty(\mathfrak{g}^*)$.

(ii) Everything is in Chapter 3. The source of a symplectic groupoid is *anti*-Poisson onto the base structure (Theorem 3.1), and the base structure of the cotangent groupoid is the Lie–Poisson bracket π_+ of Definition 2.4 (Example 3.3, recovered independently by Theorem 3.2). Therefore

$$\{s^*f, s^*g\}_\Omega = -s^*\{f, g\}_+ = s^*\{f, g\}_-.$$

For the dynamics, anti-Poissonness applied to Hamiltonian vector fields gives $Ts(X_{s^*h}) = -X_h^{\pi_+} \circ s = X_h^{\pi_-} \circ s$. The canonical flow of s^*h therefore projects along s to the π_- -flow of h . On $\mathfrak{so}(3)^*$ the bracket π_- is $\{f, g\}(\boldsymbol{\mu}) = -\boldsymbol{\mu} \cdot (\nabla f \times \nabla g)$, and for $h = \frac{1}{2} \boldsymbol{\mu} \cdot \mathbb{I}^{-1} \boldsymbol{\mu}$ the resulting equation is $\dot{\boldsymbol{\mu}} = -\nabla h \times \boldsymbol{\mu} = \boldsymbol{\mu} \times \boldsymbol{\Omega}$. $\square$

The space-frame description is the mirror image. Trivialising by right translations records the spatial momentum $\boldsymbol{m} = (T_e R_g)^* \alpha_g = \text{Ad}_g^* \mu = t(g, \mu)$, and its functions close on π_+ because the target is a Poisson map (Theorem 3.1).

The map that exchanges the two descriptions, $(g, \mu) \mapsto (g^{-1}, \text{Ad}_g^* \mu)$, is the groupoid inversion ι , and $\iota^* \Omega = -\Omega$ by Proposition ???. That single sign is the sign separating π_- from π_+ . So the two Poisson structures on $\mathfrak{g}^*$, left unexplained in Chapter 2, are the two frames of the mechanics, and switching frame means inverting arrows.

Momentum maps are the structure maps

Chapter 2 introduced momentum maps abstractly (Definition ??), as maps $J: M \rightarrow \mathfrak{g}^*$ whose components generate the symmetry. For the two translation actions on T^*G the momentum maps are classical [8], and they turn out to be objects we already have. The mechanism is simple. Both lifted actions preserve not only Ω but its primitive θ , because cotangent lifts always preserve the tautological one-form. And a flow that preserves θ is automatically Hamiltonian. If V generates it, then Cartan's formula gives

$$0 = \mathcal{L}_V \theta = \iota_V d\theta + d(\theta(V)) \implies \iota_V \Omega = d(\theta(V)),$$

so $V = X_{\theta(V)}$, with the Hamiltonian function $\theta(V)$ read off from the tautological form.

Proposition 4.2. *On $T^*G \cong G \times \mathfrak{g}^*$, the lifted left-translation action (4.2.1) is Hamiltonian with equivariant momentum map the target,*

$$J_L = t, \quad \langle J_L(g, \mu), \xi \rangle = \langle \text{Ad}_g^* \mu, \xi \rangle,$$

and the lifted right-translation action, $(g, \mu) \cdot h = (gh, \text{Ad}_{h^{-1}}^ \mu)$, is Hamiltonian with momentum map the source, $J_R = s$.*

Proof. The generator of (4.2.1) in the direction $\xi \in \mathfrak{g}$ is $V_\xi(g, \mu) = \frac{d}{d\tau} \big|_0 (\exp(\tau\xi)g, \mu) = (T_e R_g \xi, 0)$. By the computation above, V_ξ is Hamiltonian with function

$$\theta(V_\xi) = \langle \mu, \theta^L(T_e R_g \xi) \rangle = \langle \mu, \text{Ad}_{g^{-1}} \xi \rangle = \langle \text{Ad}_g^* \mu, \xi \rangle,$$

which is the ξ -component of t . Equivariance is the identity $t(h \cdot (g, \mu)) = \text{Ad}_{hg}^* \mu = \text{Ad}_h^* t(g, \mu)$. For right translations, the same trivialisation computation as for (4.2.1) gives the lift $(g, \mu) \cdot h = (gh, \text{Ad}_{h^{-1}}^* \mu)$, with generator $W_\xi(g, \mu) = (T_e L_g \xi, \text{ad}_\xi^* \mu)$ and $\theta(W_\xi) = \langle \mu, \xi \rangle$, the ξ -component of s . The handedness of a right action brings the usual reversed equivariance. It is the same bookkeeping as the sign in Theorem 4.1. $\square$

Noether's theorem now costs one line. The free rigid body has a left-invariant Hamiltonian, $H = s^* h$ with $h(\mu) = \frac{1}{2} \mu \cdot \mathbb{I}^{-1} \mu$, so the conserved quantity of the symmetry should be $J_L = t$, the spatial momentum. Indeed, for any function φ on $\mathfrak{g}^*$,

$$\frac{d}{d\tau} (t^* \varphi) = \{t^* \varphi, s^* h\}_\Omega = 0,$$

because source pullbacks Poisson-commute with target pullbacks. That identity was the first display in the proof of Theorem 3.1. Conservation of spatial angular momentum is, on the groupoid, the statement that the two legs of the dual pair of Remark ?? commute. This is also consistent with Proposition ??, since t is a Poisson map onto $(\mathfrak{g}^*, \pi_+)$.

Flows as paths of bisections

Section 4.1 pictured a Hamiltonian flow as a path of Lagrangian bisections. We want the same picture over the degenerate base, and Remark 4.1 warned that it is not automatic. On $\mathfrak{g}^*$, a Hamiltonian flow φ_t preserves each symplectic leaf, so for each μ *some* group element carries μ to $\varphi_t(\mu)$. But a bisection is a single smooth section, not a pointwise choice, and the group element is not unique. The following construction resolves this. Instead of choosing group elements point by point, one flows the whole unit section inside the groupoid. The statement is classical and goes back to Coste, Dazord and Weinstein [10]. Its proof is short and instructive, so we include it.

Proposition 4.3. *Let $(\mathcal{G} \rightrightarrows M, \Omega)$ be a symplectic groupoid over (M, π) as in Theorem 3.1, let $H \in C^\infty(M)$ have Hamiltonian flow φ_t on M , and let $\widehat{\varphi}_\tau$ denote the flow on $\mathcal{G}$ of the Hamiltonian vector field X_{s^*H} . For every t for which $\widehat{\varphi}_t$ is defined on all of $u(M)$, the set*

$$B_t := \widehat{\varphi}_t(u(M))$$

is a Lagrangian bisection of $\mathcal{G}$, and its induced base map is the flow: $\psi_{B_t} = \varphi_t$.

Proof. There are three things to check.

B_t is Lagrangian. The map $\widehat{\varphi}_t$ is the flow of a Hamiltonian vector field on $(\mathcal{G}, \Omega)$, hence a symplectomorphism where defined. The unit section is Lagrangian (Proposition 3.2), and the image of a Lagrangian submanifold under a symplectomorphism is Lagrangian.

B_t is a bisection. The proof of Theorem 3.1 showed $X_{s^*H} \in \ker Tt$, so the flow does not move targets: $t \circ \widehat{\varphi}_t = t$. Hence $t \circ (\widehat{\varphi}_t \circ u) = t \circ u = \text{id}_M$, and t restricts to a diffeomorphism $B_t \rightarrow M$. For the source, we claim $s \circ \widehat{\varphi}_t = \varphi_{-t} \circ s$. Indeed, for every $f \in C^\infty(M)$,

$$df(Ts(X_{s^*H})) = X_{s^*H}(s^*f) = \{s^*f, s^*H\}_\Omega = -s^*\{f, H\} = -(X_H f) \circ s,$$

using that s is anti-Poisson (Theorem 3.1). So X_{s^*H} projects along s to $-X_H$, and its flow projects to φ_{-t} . Consequently $s \circ (\widehat{\varphi}_t \circ u) = \varphi_{-t} \circ s$, a diffeomorphism of M , and s also restricts to a diffeomorphism $B_t \rightarrow M$.

The base map is φ_t . From the last display, $(s|_{B_t})^{-1} = (\widehat{\varphi}_t \circ u) \circ \varphi_t$, and therefore

$$\psi_{B_t} = t|_{B_t} \circ (s|_{B_t})^{-1} = (t \circ \widehat{\varphi}_t \circ u) \circ \varphi_t = \text{id}_M \circ \varphi_t = \varphi_t. \quad \square$$

On the pair groupoid the construction reproduces $B_t = \text{Gr}(\varphi_t)$, so the picture of Section 4.1 holds over any Poisson base.

For the cotangent groupoid the proposition becomes very concrete. Since $s|_{B_t}$ is a diffeomorphism, B_t is the image of a smooth section $\mu \mapsto (b_t(\mu), \mu)$ of the source, with $b_t: \mathfrak{g}^* \rightarrow G$ and $b_0 \equiv e$, and the identity $\psi_{B_t} = \varphi_t$ reads

$$\text{Ad}_{b_t(\mu)}^* \mu = \varphi_t(\mu). \quad (4.2.2)$$

In words, *the exact flow of any Hamiltonian on $\mathfrak{g}^*$ moves each momentum by a coadjoint transformation, with a smoothly chosen group element.*

Trajectories can never leave their coadjoint orbit, which gives a second, dynamical view of Theorem 2.2. And the conservation of all Casimirs at once is written into the form of (4.2.2), not into the particular Hamiltonian. Equation (4.2.2) is what the next section discretises.

4.3. Poisson integrators

Most differential equations cannot be solved in closed form, and one computes trajectories numerically. This section applies the groupoid picture to that task. It starts from zero. The few notions of numerical analysis that we need are introduced as we go, and the method is built in stages. Each stage adds one idea, and we state plainly what the idea buys.

First, the vocabulary. A *one-step method* for an equation $\dot{x} = f(x)$ replaces the unknown flow φ_t by a computable map Φ_h , the *update*, applied once per time step h :

$$x_{n+1} = \Phi_h(x_n), \quad x_n \approx \varphi_{nh}(x_0).$$

The simplest update follows the tangent line for time h ,

$$\Phi_h(x) = x + h f(x),$$

the *explicit Euler method*. A method has *order p* if the error after a fixed time T shrinks like h^p as the step is refined. Halving the step divides the error by about 2^p . Euler has order one. Order measures short-time accuracy. For long simulations a second question matters just as

much. Which properties of the exact flow does Φ_h share? Errors accumulate over thousands of steps, and whether they accumulate in a harmless or a destructive way depends on the shared structure. This question is the subject of *geometric numerical integration* [17], and Lie–Poisson systems have been among its central examples from the start [18, 20].

The route through this section is short to state. We first watch the simplest method fail on the simplest spin system, and we measure the failure. We then change the *form* of the update, guided by (4.2.2), and the failure disappears for every Casimir at once. Symmetry raises the accuracy to second order. A theorem of Ge and Marsden then marks the limit of what any method can preserve. Finally, the generating function of Section 3.6 upgrades the construction to the bracket itself, and to Poisson manifolds far beyond $\mathfrak{g}^*$.

A first test: spin precession

We take the simplest Lie–Poisson system there is. On $\mathfrak{so}(3)^* \cong \mathbb{R}^3$ with the bracket of Example ??, let the Hamiltonian be linear, $H(\boldsymbol{\mu}) = \mathbf{B} \cdot \boldsymbol{\mu}$ for a constant vector $\mathbf{B}$. The equation of motion $\dot{\boldsymbol{\mu}} = \nabla H \times \boldsymbol{\mu}$ becomes

$$\dot{\boldsymbol{\mu}} = \mathbf{B} \times \boldsymbol{\mu},$$

which the reader will recognise. After absorbing physical constants, this is the precession of a spin or magnetic moment $\boldsymbol{\mu}$ in a constant field $\mathbf{B}$. The exact solution is a rotation at constant rate about the axis $\mathbf{B}$,

$$\boldsymbol{\mu}(\tau) = \exp(\tau \hat{\mathbf{B}}) \boldsymbol{\mu}_0,$$

where $\hat{\mathbf{B}}$ is the skew matrix with $\hat{\mathbf{B}}\mathbf{v} = \mathbf{B} \times \mathbf{v}$. Two facts are clear from this formula. The length $|\boldsymbol{\mu}|$ is constant, and the motion stays on a sphere. The first is the Casimir of Example ??, the second is the symplectic leaf. Any credible simulation of a precessing spin should keep the spin length.

Explicit Euler does not, and the failure can be computed in two lines, for any Hamiltonian on $\mathfrak{so}(3)^*$.

Lemma 4.1. *For a Hamiltonian H on $\mathfrak{so}(3)^*$, the explicit Euler method $\boldsymbol{\mu}_{n+1} = \boldsymbol{\mu}_n + h \nabla H_n \times \boldsymbol{\mu}_n$ satisfies*

$$|\boldsymbol{\mu}_{n+1}|^2 = |\boldsymbol{\mu}_n|^2 + h^2 |\nabla H_n \times \boldsymbol{\mu}_n|^2.$$

The Casimir $|\boldsymbol{\mu}|^2$ grows at every step at which the right-hand side of the equation of motion is non-zero. Refining h slows the growth but never removes it.

Proof. The increment $h \nabla H_n \times \boldsymbol{\mu}_n$ is orthogonal to $\boldsymbol{\mu}_n$, so the cross terms in the squared norm vanish. $\square$

For the precessing spin this means that $|\boldsymbol{\mu}|$ increases at every single step. The simulated spin spirals outward through the spheres and its magnitude grows without bound, which is physically absurd. The exact motion rotates the sphere within itself, while the Euler step leaves along the tangent line and lands outside the sphere. That standard numerical methods fail in this way on Lie–Poisson systems is well documented [17, 20]. The lemma above makes the failure quantitative in the simplest case.

Could a more clever classical method repair this? Partly, and the partial success is instructive. Certain implicit methods, such as the implicit midpoint rule, conserve all *quadratic* conserved quantities exactly [17], and $|\boldsymbol{\mu}|^2$ is quadratic, so they would keep our sphere. But this is an accident of degree. The Casimir of $\mathfrak{so}(3)^*$ happens to be quadratic. Other algebras have cubic Casimirs, $\mathfrak{su}(3)^*$ is the standard example, and a classical result says that no Runge–Kutta method can conserve all invariants of degree three [17]. The lesson is this.

Conservation of the Casimirs, if we want it for every algebra and every Hamiltonian, cannot be obtained by tuning the coefficients of a method built from straight-line steps. It must be built into the *form* of the update.

The idea: step with the group

For the precessing spin the right form is easy to see. Do not translate. Rotate:

$$\mu_{n+1} = \exp(h\hat{B}) \mu_n.$$

For a constant field this is not merely a better approximation. It is the exact flow at time h , and the spin length is conserved because rotations preserve length.

The groupoid turns this observation into a principle, because Section 4.2 showed what the exact flow of *any* Hamiltonian on $\mathfrak{g}^*$ looks like. By (4.2.2) it moves each momentum by a coadjoint transformation, $\varphi_t(\mu) = \text{Ad}_{b_t(\mu)}^* \mu$, with a smooth family of group elements coming from a Lagrangian bisection. A numerical method should imitate the form and approximate only the group element. We therefore consider updates

$$\Phi_h(\mu) = \text{Ad}_{c_h(\mu)}^* \mu, \quad (4.3.1)$$

where $\mu \mapsto (c_h(\mu), \mu)$ is a smooth section of the source with $c_0 \equiv e$, that is, a bisection of the cotangent groupoid for each h . We call these *bisection schemes*. Updates of this form are the Lie–Poisson case of the Lie-group methods of numerical analysis [23, 21], and the idea of reading them through the symplectic groupoid goes back to Ge [19]. The point of the definition is the following observation, which is the common reason why all such methods work.

Proposition 4.4. *Every method of the form (4.3.1) keeps each coadjoint orbit invariant, and therefore conserves every Casimir of $(\mathfrak{g}^*, \pi_\pm)$ exactly: $C(\Phi_h(\mu)) = C(\mu)$ for all μ , all h and every Casimir C , no matter how inaccurate c_h is.*

Proof. The point $\text{Ad}_{c_h(\mu)}^* \mu$ lies in the coadjoint orbit of μ by construction. The orbits are the symplectic leaves (Theorem 2.2), and Casimirs are constant on leaves (Theorem 2.1). $\square$

This is the first payoff, and it deserves plain words. In a bisection scheme, two questions that classical numerical analysis had to handle together come apart. *Where the step is allowed to go* is decided by the form (4.3.1), once and for all. The step never leaves the orbit, so all Casimirs are exact. *How accurately it goes there* is a separate question, and the only one where approximation enters, through c_h . In exact arithmetic the conservation is exact. On a computer it holds to rounding error, uniformly in h and in the number of steps, because it is an identity of the map and not a cancellation of errors. What the form does *not* give by itself is the second level of structure. A bisection need not be Lagrangian, so Φ_h need not preserve the Poisson bracket. We come back to this at the end of the section. The two levels separate along the two halves of Remark 4.1.

The rigid body: a first working scheme

We now let the Hamiltonian be nonlinear, and take the free rigid body of Section 4.2, in the π_+ convention of Example ??: $\dot{\mu} = \Omega \times \mu$ with $\Omega = \mathbb{I}^{-1}\mu$. It is the running example of this thesis, and also the standard test problem of the literature below. Its exact solution exists but involves elliptic functions, so a numerical method is genuinely useful. The new difficulty compared with precession is that the rotation axis now depends on the point. The body precesses about an axis that moves with it.

The simplest bisection scheme freezes the axis over one step. Guided by (4.2.2), we take $c_h(\mu) = \exp(h \, d_\mu H)$, which for the rigid body means one rotation about the current angular velocity:

$$\Phi_h(\mu) = \text{Ad}_{\exp(h \, d_\mu H)}^* \mu, \quad \mu_{n+1} = \exp(h \, \widehat{\Omega}_n) \mu_n. \quad (4.3.2)$$

Schemes of this type appear in the work of Engø and Faltinsen [23], within a family of arbitrary order, and explicit relatives were built by McLachlan [21]. The cost per step is one evaluation of dH and one exponential. For $\text{SO}(3)$ the exponential is the explicit Rodrigues rotation formula, so a step is a few lines of code. How accurate is it? Expanding with $\frac{d}{d\tau} \big|_0 \text{Ad}_{\exp(\tau\xi)}^* \mu = -\text{ad}_\xi^* \mu$ gives

$$\Phi_h(\mu) = \mu - h \, \text{ad}_{d_\mu H}^* \mu + O(h^2) = \mu + h \, X_H(\mu) + O(h^2),$$

which matches the Lie–Poisson equation (2.3.1) to first order. So the scheme has order one, the same as explicit Euler, and yet by Proposition 4.4 it conserves every Casimir exactly, while Euler drifts at every step. Same accuracy, different geometry. This is the cleanest illustration of what the form buys.

Second order from symmetry

The scheme (4.3.2) is lopsided. It reads the axis at the start of the step, while the exact flow has no preferred time direction. A method that treats the two ends of a step on an equal footing is called *symmetric*, and a standard theorem says that symmetric methods automatically have even order [17]. The natural symmetric variant of (4.3.2) reads the axis at the midpoint of the rotation. For the rigid body:

$$\mu_{n+1} = \exp(h \, \widehat{\Omega}(\nu)) \mu_n, \quad \text{where } \nu = \exp\left(\frac{h}{2} \widehat{\Omega}(\nu)\right) \mu_n. \quad (4.3.3)$$

Closely related implicit schemes appear in [23]. The auxiliary point ν is defined implicitly, but for small h its equation has a unique solution near μ_n , and simple fixed-point iteration finds it. The cost per step stays at one or two exponentials. Let us check the symmetry. Splitting the full rotation into two halves gives $\mu_{n+1} = \exp\left(\frac{h}{2} \widehat{\Omega}(\nu)\right) \nu$, so

$$\nu = \exp\left(\frac{h}{2} \widehat{\Omega}(\nu)\right) \mu_n = \exp\left(-\frac{h}{2} \widehat{\Omega}(\nu)\right) \mu_{n+1}.$$

The point ν is the true midpoint of the rotation, seen identically from both ends, and exchanging $(\mu_n, h) \leftrightarrow (\mu_{n+1}, -h)$ reproduces the same pair of relations. That is the definition of a symmetric method. The scheme is also consistent, since $\nu = \mu_n + O(h)$ reduces it to (4.3.2) at leading order. Being symmetric and consistent, it has order two [17]. Being a single rotation, it conserves all Casimirs exactly, by Proposition 4.4 again. The same two lines work on any $\mathfrak{g}^*$, with $\nu = \text{Ad}_{\exp((h/2)d_\nu H)}^* \mu_n$ and the full step $\text{Ad}_{\exp(h \, d_\nu H)}^*$.

Remark 4.2. Readers who know the adjoint method $\Phi_h^* = (\Phi_{-h})^{-1}$ of numerical analysis [17] may enjoy the groupoid translation. The inverse of a bisection in the bisection group is its image under the inversion ι (Definition 3.3 and the discussion following it). Passing to the adjoint therefore replaces the bisections C_h of a scheme by $\iota(C_{-h})$, and symmetric schemes are the ones aligned with the inversion. Since ι is anti-symplectic (Proposition ??), it sends Lagrangian bisections to Lagrangian bisections. The map that exchanged the body and space frames in Section 4.2 is here doing numerical work.

What cannot be preserved: the energy

Neither (4.3.2) nor (4.3.3) conserves the Hamiltonian exactly, and one might hope to fix this next. A theorem of Ge and Marsden says the hope is misplaced. We state it informally. The hypotheses are made precise in [18].

Theorem 4.2 (Ge–Marsden). *Suppose a one-step method with fixed step size is a Poisson map and conserves the Hamiltonian exactly, and suppose the system has no conserved quantities other than functions of the Hamiltonian. Then the method is the exact flow, up to a change of the time variable.*

In other words, a method that is not secretly the exact solution cannot have everything. Of the three structures in play, Casimirs, bracket and energy, at most two can be exact. Bisection schemes take the Casimirs for free and, as we will see next, can take the bracket. The theorem then says the energy must stay approximate. The choice can also be made the other way. Engø and Faltinsen [23] give orbit-preserving schemes that conserve the energy exactly, and, consistently with the theorem, those schemes are not Poisson maps.

How bad is an approximate energy? Much less bad than Lemma 4.1 would suggest, and the reason is the central result of backward error analysis [17]. For a structure-preserving method of order p , the computed points lie, to very high accuracy and over very long times, on exact trajectories of a *modified* Hamiltonian $H_h = H + O(h^p)$. The numerical energy therefore does not drift away. It oscillates in a band of width $O(h^p)$ around its initial value, for as long as one cares to integrate. Compare the two failure modes. Euler’s Casimir error grows at every step at a computable rate, while the energy error of a structure-preserving method stays bounded. For bisection schemes there is a refinement that keeps our geometry in view. The modified Hamiltonian can be constructed on the groupoid itself, so the modified system is Hamiltonian for the *same* Poisson structure, with the same leaves and the same Casimirs, and only the energy function is perturbed [27].

The last structure: the bracket

One level of structure remains. The update should be a Poisson map, so that it preserves the symplectic area on each leaf and not only the leaf itself. The two ingredients are already on the table. Proposition 3.6 says that Lagrangian bisections induce Poisson maps. Proposition 4.3 says that the exact flow is a path of Lagrangian bisections. So the task is concrete. Find bisections that are Lagrangian, close to those of the exact flow, and computable.

Being Lagrangian is a condition on derivatives, and such conditions are solved by potentials. The model case is classical mechanics, where a canonical transformation near the identity is encoded by a single generating function [8]. The converse is the useful part. Any function of this kind generates a canonical transformation, because a function, unlike a map, has no condition left to satisfy. The same mechanism runs on a symplectic groupoid. Near the unit section, one scalar function encodes one Lagrangian bisection, and any nearby function gives one. This is exactly the device of Section 3.6, where Karasev and Weinstein built the local symplectic groupoid out of a generating function solving a Hamilton–Jacobi equation [9, 11].

Now the strategy almost writes itself. The Lagrangian bisections B_h of the exact flow correspond to a generating function S_h , and S_h expands in powers of the step, with the Hamiltonian as the term linear in h [18]. The exact S_h is out of reach, because knowing it would mean knowing the flow. But its Taylor polynomial in h can be computed term by term from H and the bracket. And here is the key point. A truncated generating function is still a function. So it still generates an honest Lagrangian bisection, whose base map is an *exact* Poisson map that follows the true flow to the order of the truncation. Nothing about the bracket is approximated. Only the motion is. Methods built this way are called *Poisson integrators*. Their Casimirs are exact by Proposition 4.4, their bracket is exact by Proposition 3.6, and their energy follows a modified Hamiltonian, as in the previous

subsection. This is the sense in which, as Section 3.6 announced, one step of Karasev’s generating function becomes one step of a numerical scheme.

The programme took time to mature. Ge and Marsden formulated it in 1988 for Lie–Poisson spaces [18], Ge developed the groupoid picture behind it [19], and Channell and Scovel turned it into working algorithms and numerical experiments [20]. The Hamilton–Jacobi side of the theory was developed in [26]. The general case, a Poisson integrator of any order on any Poisson manifold, was completed only recently by Cosserat [27], with tests on mechanical examples, the rigid body among them, in [28]. We take the order and convergence theory from these references and do not reproduce it.

Remark 4.3. Global integrability, in the sense of Section 3.6, is not needed for any of this. A numerical step only ever uses arrows near the unit section, and a *local* symplectic groupoid over a neighbourhood of the units exists for every Poisson manifold, by the constructions of Karasev and Weinstein just cited [11, 9]. Even the non-integrable Poisson manifold of Example 3.9, which admits no global symplectic groupoid, supports groupoid-based Poisson integrators. Where the obstruction theory of Chapter 3 closes a door for global questions, the numerical application walks through the local one.

Where these methods matter

The rigid body has carried the exposition, so it is fair to ask where else the theory applies. The honest answer has two parts. Which systems, and which situations.

Which systems? Degenerate brackets are not exotic. They appear whenever a symmetry has been used to shrink a phase space, which is what Section 4.2 did, so every reduced system is a candidate. The heavy top, a spinning top with gravity acting on it, has a Lie–Poisson formulation on the dual of the Euclidean algebra $\mathfrak{se}(3)^*$, where the Casimirs again pick out the physical leaves [8]. Satellite attitude dynamics lives on $\mathfrak{so}(3)^*$, our running example. A classical spin chain has one sphere per site, so its phase space is a product of coadjoint orbits. Further out, the Euler equations of an ideal fluid and the standard models of plasma physics are Lie–Poisson systems for suitable infinite-dimensional algebras [8], and making discretisations respect that structure is a research field of its own. We only point at it here. Closest to this thesis, the generating-function construction is not tied to Lie–Poisson brackets at all. It runs on the local groupoid of *any* Poisson manifold, including brackets whose coefficients vary from point to point [27, 30]. That is the widest boundary of the method, and it is the direct numerical payoff of the local integrability discussed in Chapter 3.

Which situations? Structure preservation is a long-time property, and it is in long runs that it pays. For a short trajectory computed to high accuracy, a standard high-order method is simpler and entirely adequate [17]. The picture changes when the run is long and the question is qualitative. Think of a satellite simulated over years of flight, millions of time steps. With explicit Euler the spin length grows at every single step, by Lemma 4.1, and the model satellite slowly tumbles away from reality. With a bisection scheme the spin length is exact for as long as the program runs, and the energy stays in a thin band around its true value. The cost per step is comparable, one or two matrix exponentials. What the geometry buys is not speed. It is the right to trust a long computation.

Recent developments

The circle of ideas presented here has become an active subfield in the last few years, and its working vocabulary is the vocabulary of this thesis. Rather than survey it, we look at

two developments in a little more detail. They are chosen because the reader of this thesis already owns every concept they use.

The first asks what a Poisson integrator really needs. Cabrera, Martín de Diego and Vaquero [30] build their methods not on a symplectic groupoid but on what they call realisation data. That data consists of two surjections from a symplectic manifold, one Poisson and one anti-Poisson, with symplectically orthogonal fibres, together with a Lagrangian section playing the role of the units. This is word for word the structure that Chapter 3 extracted from a symplectic groupoid, in Theorem 3.1, Lemma ?? and Remark ?. A groupoid multiplication is not part of the input. So the parts of the groupoid that the numerics needs are exactly the parts that the structure theory of Chapter 3 proved to exist. The same authors show that the generating function of the exact flow is an *odd* function of the step. Through the correspondence between functions and bisections, this oddness is the inversion symmetry of Remark 4.2, now visible at the level of a single function, and it explains why the exact flow, read as a one-step method, is symmetric.

The second development steps around $\mathfrak{g}^*$ rather than on it. In the collective integrators of McLachlan, Modin and Verdier [25], one lifts the Hamiltonian through a Poisson map from a symplectic manifold, for us the target map t of Theorem 3.1, runs an ordinary symplectic method upstairs, and pushes the computed points back down. Under conditions on the method spelled out in [25], the projected update is a Poisson map. The geometry that makes the lift possible is the symplectic realisation recorded in Remark ?. In effect, the two developments use the two legs of the same dual pair.

We end with signposts. There is now a calculus of trees, in the style of Butcher’s theory for Runge–Kutta methods, that organises the order conditions of Poisson integrators on groupoids [31]. There are experiments in solving the Hamilton–Jacobi equation on the local groupoid by machine learning, where the geometry guarantees that whatever function the optimisation returns, the resulting scheme still conserves every Casimir [29]. And on the Lagrangian side of mechanics there is a parallel theory of variational integrators on Lie groupoids, begun by Weinstein [22] and developed by Marrero, Martín de Diego and Martínez [24], which we do not enter.

So what has been gained? A Poisson manifold on its own gives a numerical method little to hold on to. The bracket is degenerate, and there is no larger symplectic space in sight. The symplectic groupoid is the object with just enough room. It is large enough to carry a non-degenerate form, and structured enough to remember which Poisson manifold it serves. On it, a time step becomes a bisection, conservation of Casimirs becomes an identity, and the Poisson property becomes a Lagrangian condition, made computable by generating functions. The one structure that cannot also be exact is identified by Theorem 4.2 and controlled by a modified Hamiltonian on the same groupoid. That an object introduced in the 1980s for the quantisation of Poisson manifolds now organises their numerical analysis is a good measure of how natural the notion is. The Conclusion takes stock of both directions.

Conclusion

This thesis walked a single chain. It went from Hamiltonian mechanics to Poisson manifolds, from a Poisson manifold to its cotangent Lie algebroid, and from the algebroid to the symplectic groupoid that integrates it. Chapter 1 supplied the language of Lie groupoids and Lie algebroids. Chapter 2 supplied the Poisson geometry and built the cotangent algebroid. Chapter 3 joined the two ends, and Chapter 4 spent the result on mechanics.

The lesson we would single out is how much followed from how little. A symplectic groupoid is defined by one compatibility condition, the multiplicativity of the symplectic form. Chapter 3 drew from it, one piece at a time, the Lagrangian units, the anti-symplectic inversion, the dual pair formed by source and target, the Poisson structure of the base, and the identification of the Lie algebroid of the groupoid with the cotangent algebroid of the base. Nothing had to be added by hand. The structure was inside the definition from the start.

The analogy with Lie theory that guided the construction breaks exactly once, at integrability. A Poisson manifold, unlike a Lie algebra, may integrate to no groupoid at all, and Example 3.9 showed the failure on a small deformation of the rigid body bracket. Locally, however, the integration always exists, and Chapter 4 showed that the local object already carries the applications.

On the applied side, three things were gained. Canonical transformations and Hamiltonian flows received their groupoid names in Section 4.1. Lie–Poisson reduction of the rigid body, usually proved by a coordinate computation, became a corollary of the structure theory, with the body and space momenta appearing as the source and the target of the cotangent groupoid (Section 4.2). And Section 4.3 used the same geometry to design numerical methods. There, a time step is a bisection, and the Casimirs are exact because of the form of the update alone. The bracket is exact when the bisection is Lagrangian, and the theorem of Ge and Marsden marks the one structure, the energy, that must remain approximate.

The thesis is an exposition, and its boundaries should be stated as plainly as its results. The integrability theorem of Crainic and Fernandes was used, not proved [14]. The Hamilton–Jacobi machinery behind higher-order Poisson integrators was quoted from the literature, not reproduced. Infinite-dimensional systems, fluids among them, were only pointed at. And everything here is finite-dimensional, smooth and classical.

Two directions continue the story, and they were the two motivations all along. The first is numerical, and it is unfolding now. The final pages of Section 4.3 recorded a field in motion, with the minimal input for a Poisson integrator being isolated, order conditions being organised into trees, and generating functions being learned by optimisation. A natural continuation of this thesis, at master level, would be concrete. Implement a generating-function integrator on the local groupoid of some non-linear Poisson bracket, the deformed rigid body bracket of Example 3.9 being a tempting candidate, and test the theory against the standard methods.

The second direction is the one the founders had in mind. Karasev, Weinstein and

Zakrzewski introduced symplectic groupoids as the classical skeletons of hoped-for quantum algebras [11, 9, 12]. In that programme the groupoid is not a tool for computing trajectories. It is the object one quantises. The aim, in outline, is to apply geometric quantisation to $(\mathcal{G}, \Omega)$ in a way that respects the multiplication, and to obtain an algebra of operators that deforms the functions on the Poisson base [4]. A related route runs through field theory. The Poisson sigma model of Cattaneo and Felder is a two-dimensional field theory attached to any Poisson manifold, and its reduced phase space turns out to be precisely the symplectic groupoid, when the groupoid exists [16]. This places the object of this thesis at the door of deformation quantisation. Walking through that door needs tools beyond a bachelor thesis, which is exactly why it makes a horizon.

The free rigid body accompanied us from Chapter 2 to the last numerical scheme, and it is a fair summary of the thesis that one small physical system, looked at patiently, forced the whole theory into view. The symplectic groupoid began as an answer to a quantisation question that it has not yet fully answered. On the way it has organised classical mechanics, explained reduction, and started to run on computers. For an object defined by a single condition, that is a good career so far.

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